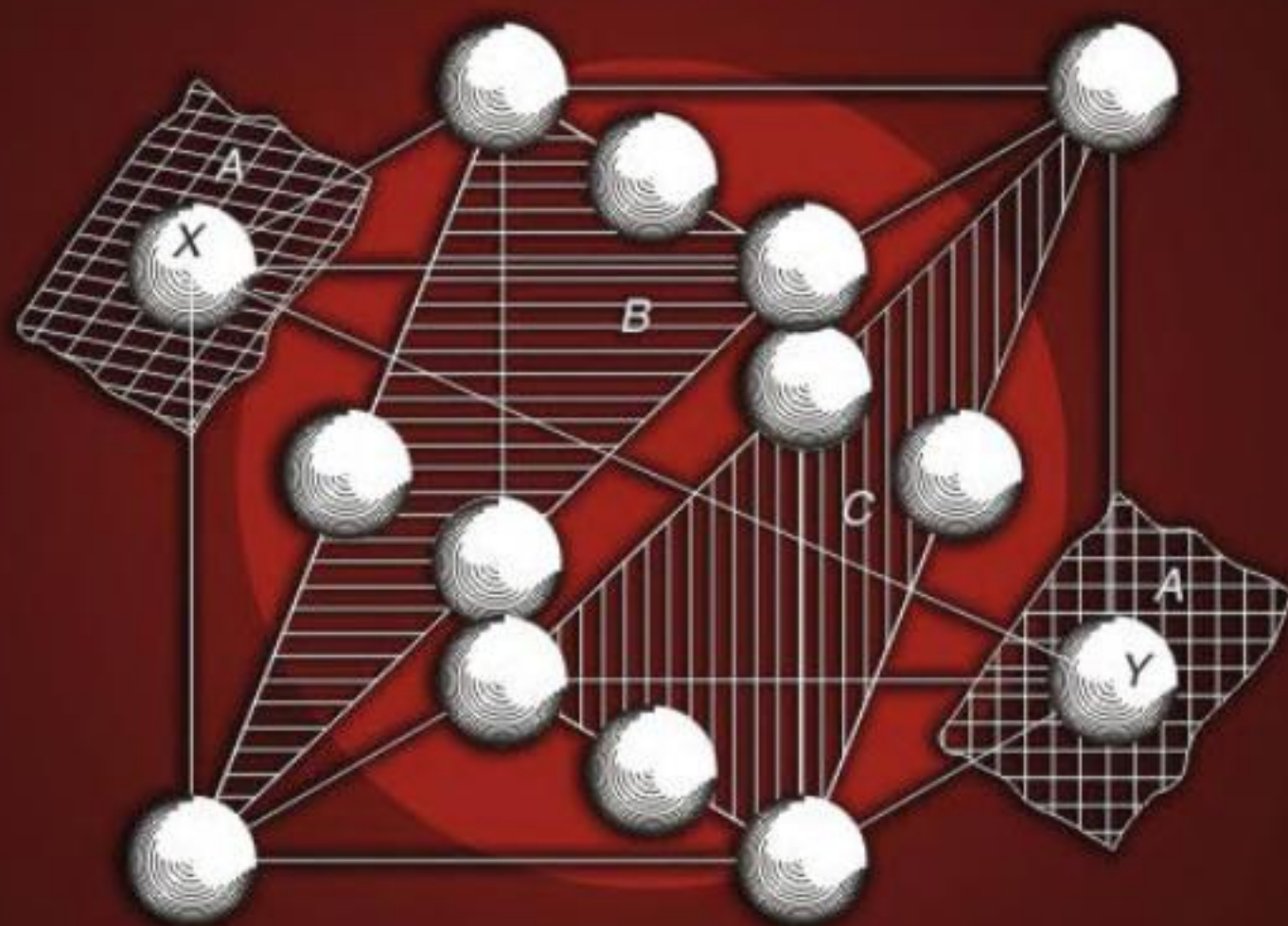


Eastern
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SECOND EDITION



PHYSICAL METALLURGY

PRINCIPLES AND PRACTICE

V. RAGHAVAN



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PHYSICAL METALLURGY: Principles and Practice, Second Edition

V. Raghavan

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PREFACE

In the Second Edition of this book, several new sections such as thermoelastic martensite, shape memory alloys, rapid solidification processing and quaternary phase diagrams have been included at appropriate places. The text has been updated and rewritten for greater clarity. The quality of diagrams has been improved and their numbers increased. The list of references at the end of each chapter has been updated.

I wish to thank Ms R. Malar for her assistance in preparing the diagrams. Thanks are due to the editorial and production team of the Publishers, Prentice-Hall of India, for their assistance and support.

V. RAGHAVAN

PREFACE TO THE FIRST EDITION

Physical Metallurgy deals with the relationship between structure and properties of metals and alloys. It forms a vital link in the processes of making, shaping and heat treating of metals. It thus interfaces with the other areas of metallurgy such as process metallurgy, mechanical metallurgy and engineering metallurgy. Voluminous books are available which deal with the fundamentals of this subject in a detailed and rigorous manner. In addition, there are books dealing with specific topics such as the structural aspect at an advanced level. At the other end, some books present just factual data on the properties and uses of various grades of steels and nonferrous alloys. There is thus a clear need for a book that gives a broad introduction to the subject dealing with the principles at a reasonable depth and at the same time discussing their relationship to practice. This book aims to do that.

All undergraduate students of engineering and metallurgy study phase diagrams, heat treatment principles, mechanical behaviour of alloys, corrosion phenomena, and properties and uses of specific engineering alloys. Undergraduate students of metallurgy have an introductory course on physical metallurgy in their early years. Students of engineering, specially mechanical and chemical engineering, study these topics in one course or the other—whether it be metallurgy, engineering materials or materials science and metallurgy. This book is intended for such students. In addition, the book is designed to be used by students doing self-study and preparing for the associate membership examinations of the Indian Institute of Metals (AMIIM), the Institution of Mechanical Engineers, the Institution of Engineers, etc. The simple style of the book should make it attractive to them.

The opening chapter covers the structural aspects with a discussion of the unit cells of metallic crystals, solid solutions, intermediate phases and structural imperfections. Equilibrium diagrams are introduced in the second chapter, with emphasis on the binary diagrams. Short sections on ternary systems and on experimental methods of determining phase diagrams are included. The major topics under phase changes are solidification, steel transformations and precipitation processes in nonferrous alloys. A separate chapter is devoted to the heat treatment of steels, with a detailed discussion of the concept of hardenability. The chapter on mechanical properties not only covers the basic concepts on deformation and strengthening mechanisms, but also deals with applications to creep, fatigue and fracture. Even though corrosion is a peripheral topic in physical metallurgy, in view of its importance, it is discussed in a separate

chapter. The last chapter is totally devoted to practice, giving information on structure, treatment, properties and uses of individual alloys with their specifications, covering most types of steels and common nonferrous alloys.

The author wishes to thank his colleagues in the metallurgy group of the Department of Applied Mechanics for their assistance. Support received from IIT Delhi for preparing the manuscript and the diagrams is acknowledged. Encouragement from Dr. D.P. Antia, Shri C.V. Sundaram and Professor A.K. Seal of the Indian Institute of Metals is appreciated.

New Delhi
July 20, 1983

V. RAGHAVAN

PHYSICAL CONSTANTS

Avogadro's number	$N = 6.023 \times 10^{23} \text{ mol}^{-1}$
Boltzmann's constant	$k = 1.380 \times 10^{-23} \text{ J K}^{-1}$ $= 8.614 \times 10^{-5} \text{ eV K}^{-1}$
Gas constant	$R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$
Planck's constant	$h = 6.626 \times 10^{-34} \text{ J s}$
Electronic charge	$e = 1.602 \times 10^{-19} \text{ C}$
Electron rest mass	$m_0 = 9.109 \times 10^{-31} \text{ kg}$
Velocity of light	$c = 2.998 \times 10^8 \text{ m s}^{-1}$
Bohr magneton (magnetic moment)	$\mu_B = 9.273 \times 10^{-24} \text{ A m}^2$
Permittivity of free space	$\epsilon_0 = 8.854 \times 10^{-12} \text{ F m}^{-1}$
Permeability of free space	$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$ $= 1.257 \times 10^{-6} \text{ H m}^{-1}$
Faraday's constant	$F = 96.49 \text{ kC mol}^{-1} \text{ (of electrons)}$
Atomic mass unit (amu)	$1/(10^3 N) = 1.660 \times 10^{-27} \text{ kg}$
Acceleration due to gravity	$g = 9.81 \text{ m s}^{-2}$

CONVERSION FACTORS

1 inch	=	25.4 mm
1 nm	=	10^{-9} m
1 Å	=	10^{-10} m = 0.1 nm
1°	=	1/57.3 rad
T °C	=	$(T + 273.15)$ K
T °F	=	$5/9(T + 459.67)$ K
1 per °F	=	$9/5$ K ⁻¹
1 kgf	=	9.81 N
1 lb	=	4.45 N
1 dyne	=	10^{-5} N
1 dyne/cm	=	10^{-3} N m ⁻¹
1 atmosphere	=	0.101 325 MN m ⁻²
1 bar	=	10^{-1} MPa
1 psi	=	6.89 kN m ⁻²
1 ksi (10^3 psi)	=	6.89 MN m ⁻²
1 ton/sq.in.	=	15.46 MN m ⁻²
1 kgf/cm ²	=	98.1 kN m ⁻²
1 kgf/mm ²	=	9.81 MN m ⁻²
1 dyne/cm ²	=	0.1 N m ⁻²
1 torr (mm of Hg)	=	133.3 N m ⁻²
1 kgf/mm ^{3/2}	=	0.310 MN m ^{-3/2}
1 ksi √in	=	1.10 MN m ^{-3/2}
1 eV	=	1.602×10^{-19} J
1 erg	=	10^{-7} J
1 calorie	=	4.18 J
1 eV/entity	=	96.49 kJ mol ⁻¹
1 erg/cm	=	10^{-5} J m ⁻¹
1 erg/cm ²	=	10^{-3} J m ⁻²
1 erg/cm ³	=	0.1 J m ⁻³
1 lb-in/in ²	=	175 J m ⁻²
1 lb/cu.in	=	27 680 kg m ⁻³
1 cm/cm ³	=	10^4 m m ⁻³

1 mole/cm ² /sec	=	10 ⁴ mol m ⁻² s ⁻¹
1 mole/cm ³	=	10 ⁶ mol m ⁻³
1 cm ² /sec	=	10 ⁻⁴ m ² s ⁻¹
1 mA/cm ²	=	10 A m ⁻²
1 A hr	=	3.6 kC

Chapter 1

STRUCTURE OF METALS AND ALLOYS

Physical metallurgy deals with the relationship between structure and properties of metals and alloys. The structure is influenced significantly by the process adopted for metal extraction from the ores and subsequent fabrication. The properties of interest are mainly those demanded by engineers such as strength, ductility, toughness and corrosion resistance. Thus physical metallurgy interfaces with areas of process or chemical metallurgy, mechanical metallurgy and engineering metallurgy. It provides a vital link in making, shaping, treating and using metals and alloys. In this introductory textbook, the principles and the corresponding practice of physical metallurgy are outlined in simple terms, without going into a rigorous mathematical treatment of the structure-property relationships.

1.1 NATURE OF METALLIC BONDING

The atoms in a metal are held together by chemical bonds known as *metallic bonds*. The metallic bonding has certain unique characteristics as listed below:

1. A metallic crystal can be visualized as an array of positive ions, with a common pool of electrons to which all the metal atoms have contributed their outermost electrons. The electrons in this pool have freedom to move through the crystal to any part and virtually in any direction. They are known as *free electrons*. The attractive force between the positive ions and the negatively-charged free electrons is what binds the metal atoms together in the crystal.

2. The free electrons carry both electric current and heat through the crystal. They are responsible for the *excellent electrical and thermal conductivity* of metals.
3. The bonds formed by the free electrons are *nondirectional*. There are no constraints about bond angles. An atom in a metallic crystal is bonded to as many neighbours as space permits. This provides considerable flexibility in the directions along which the metal atoms can be displaced during plastic deformation. This is mainly responsible for the unique properties of *high ductility and malleability* that metals possess. For example, gold can be drawn into extremely thin wires; aluminium can be rolled into foils less than 0.1 mm thick.

1.2 CRYSTAL STRUCTURES OF METALS

Approximately three-fourths of the elements in the periodic table are metals. More than two-thirds of them (more than 50) possess relatively simple crystal structures. Understanding these structures and their characteristics is a first step in the study of physical metallurgy.

Body centred cubic (BCC) crystal The reference unit (called the space lattice) for this crystal is a cube, with lattice points at the eight corners and at the body centre, as shown in Fig. 1.1a. In the unit cell of the crystal, atoms are positioned one at each lattice point, i.e. at the eight corners and at the body centre of the cube, as shown in Fig. 1.1b. The corner atoms are shared by other neighbouring units. As eight units meet at any corner, the share of the unit under consideration for a corner atom is one-eighth, see Fig. 1.1c. The atom at the body centre belongs entirely to the unit.



Figure 1.1 The body centred cubic crystal: (a) the BCC space lattice, (b) the unit cell of the BCC crystal, and (c) the sharing of corner atoms by the unit cell.

So, the *effective number* of atoms in the unit cell is

$$8 \text{ (corner atoms)} \times 1/8 + 1 \text{ (body centred atom)} = 2$$

The body centred atom is touching all the corner atoms. Each corner atom in turn is touching eight body centred atoms of the neighbouring cells. The

number of nearest neighbours to a given atom is called the *co-ordination number* Z . $Z = 8$ for the BCC crystal.

The distance between nearest atoms is used to define the radius r of the atom. For the BCC crystal, the centre-to-centre distance of nearest neighbours is clearly half of the body diagonal, $a\sqrt{3}/2$, where a is the length of the cube edge known as the *lattice parameter*. Noting from Fig. 1.1b that the nearest neighbours touch each other, we have

$$r = a\sqrt{3}/4$$

for the BCC crystal.

The *packing efficiency* (P.E.) of atoms in the unit cell is defined as

$$\begin{aligned} \text{P.E.} &= \frac{\text{volume of atoms in the unit cell}}{\text{volume of the unit cell}} \\ &= \frac{2 \times (4/3)\pi (a\sqrt{3}/4)^3}{a^3} \\ &= 0.68 \end{aligned}$$

The *density* ρ of the crystal can be calculated as follows :

$$\rho = \frac{\text{mass of atoms in the unit cell}}{\text{volume of the unit cell}}$$

For example, for BCC iron at room temperature,

lattice parameter $a = 2.87 \text{ \AA}$ ($2.87 \times 10^{-10} \text{ m}$)

mass of the Fe atom = 55.85 atomic mass units (amu)

$$1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$$

$$\begin{aligned} \rho_{\text{Fe}} &= \frac{2 \times 55.85 \times 1.66 \times 10^{-27}}{(2.87 \times 10^{-10})^3} \\ &= 7840 \text{ kg m}^{-3} = 7.84 \text{ g/cm}^3 \end{aligned}$$

Typical examples of metals which form BCC crystals are given below.

Metal	Cr	Fe	K	Li	Mo	Na	Nb	Ta	V	W
Lattice parameter, $\text{\AA}$	2.88	2.87	5.25	3.51	3.15	4.29	3.30	3.30	3.03	3.16

Face centred cubic (FCC) crystal The face centred cubic space lattice is shown in Fig. 1.2a. In the FCC crystal, there are eight atoms at the eight corners of the cube, with six more atoms located at the six face centres, Fig. 1.2b. A face centred atom is shared by two neighbouring unit cells, for which that face is common, as can be seen from Fig. 1.2c. The effective number of atoms in the FCC unit cell is

$$8 (\text{corner atoms}) \times 1/8 + 6 (\text{face centred atoms}) \times 1/2 = 4$$

In the FCC crystal, each atom is in contact with twelve neighbours. For example, the face centred atom of the front face in Fig. 1.2b is in contact with the four corner atoms on that face and the four face-centred atoms just behind it (on the top, the bottom and the two side faces of the unit cell). In addition, it is also in contact with four face-centred atoms of the unit cell in front of it (not seen in Fig. 1.2). So the coordination number of the FCC crystal $Z = 12$.

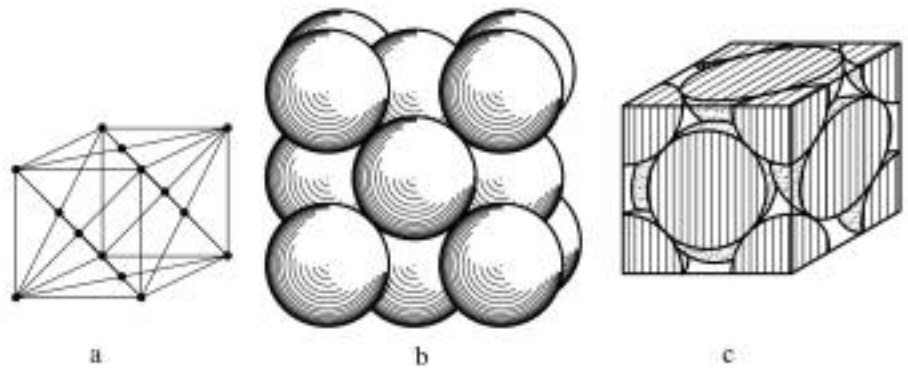


Figure 1.2 The face centred cubic crystal: (a) the FCC space lattice, (b) the unit cell of the FCC crystal, and (c) the sharing of atoms by the unit cell.

The distance between nearest neighbours is half of the face diagonal, $a\sqrt{2}/2$. The radius of the atom is $r = a\sqrt{2}/4$ for the FCC crystal.

The packing efficiency can be shown to be 0.74. Examples of metals forming the FCC crystal are listed below.

Metal	Ag	Al	Au	Cu	Fe(912-1394°C)	Ni	Pb	Pt
Lattice parameter, Å	4.09	4.05	4.08	3.61	3.65	3.52	4.95	3.92

Note that iron possesses more than one crystal form. It is BCC at low temperatures, changes over to the FCC at 912°C and back again to BCC at 1394°C. This phenomenon is called *polymorphism*.

Hexagonal close packed (HCP) crystal The hexagonal space lattice is shown in Fig. 1.3a. The atom arrangements within the HCP unit cell are shown in Fig. 1.3b. There are six atoms at the six corners of the two hexagonal faces and one each at their face centres. There are three atoms in the middle layer (where there are no lattice points). Each corner of the hexagonal prism is shared by six other prisms and the top and the bottom hexagonal faces are shared by two cells, Fig. 1.3c. The effective number of atoms in the HCP unit cell is

$$12 \text{ (corner atoms)} \times 1/6 + 2 \text{ (face centred atoms)} \times 1/2 + 3 \text{ (middle layer atoms within the cell)}^* = 6.$$

*Thin slices of the three middle-layer atoms fall outside the unit cell, but these are exactly compensated by three thin slices of outside atoms falling within the unit cell.

As in the FCC crystal, each atom here is in contact with twelve nearest neighbours. For example, referring to Fig. 1.3b, the central atom on the top hexagonal face touches the six surrounding atoms on the same face, three in the middle layer and three in the plane above (not shown), yielding $Z = 12$.

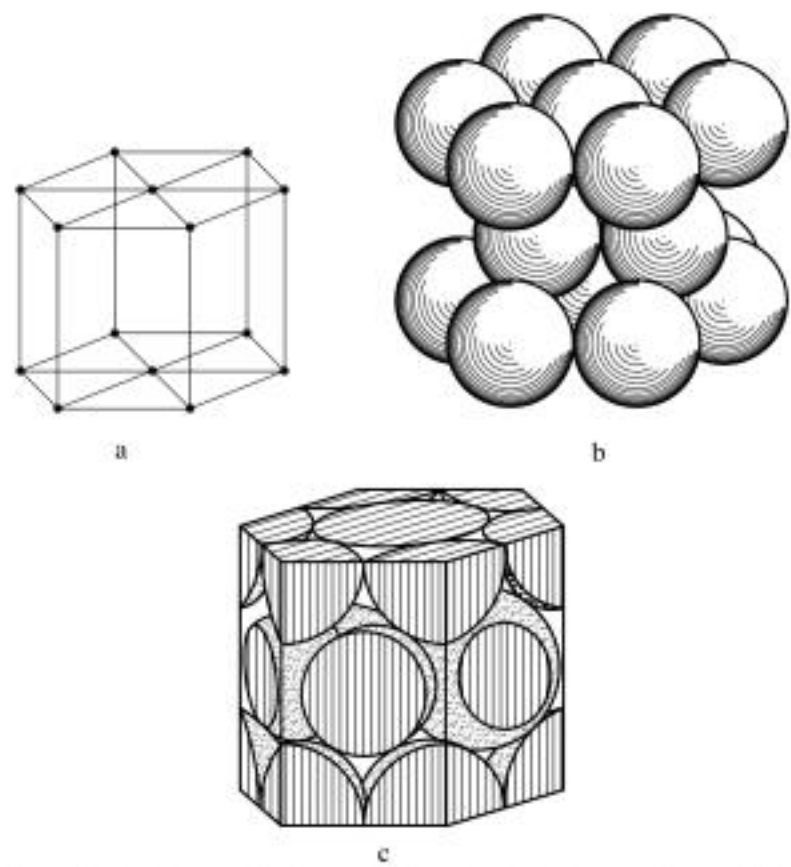


Figure 1.3 The hexagonal close packed crystal: (a) the hexagonal space lattice, (b) the HCP unit cell, and (c) the sharing of atoms by the unit cell.

The distance between nearest neighbours is the side of the hexagon a . So, $r = a/2$ for the HCP crystal.

The height of the prism gives the second lattice parameter c . The ideal packing efficiency can be shown to be the same as for the FCC crystal (0.74). Typical examples of HCP crystals and their lattice parameters are listed below.

Metal	Be	Cd	Mg	Ti (below 882°C)	Zn	Zr
a , Å	2.29	2.98	3.21	2.95	2.66	3.23
c , Å	3.58	5.62	5.21	4.68	4.95	5.15
c/a	1.56	1.89	1.62	1.59	1.86	1.59

Other less-commonly occurring metallic crystal structures include the tetragonal, the orthorhombic and the rhombohedral crystals. In the tetragonal cell, two sides of the unit cell are equal, but the third side is of a different length. In the orthorhombic unit cell, all the three sides are of different lengths. The angles between the axes in these two unit cells are the same as in the cubic crystal (90°). In the rhombohedral cell, the three axes are of the same length and the angles between axes are equal to one another, but not 90° .

Geometry of close packing For a fuller understanding of the FCC and HCP crystalline arrangements, we need to examine the geometry of close packing (meaning closest packing). Figure 1.4a shows a close packed plane *A* of equal-sized spheres. The spheres cannot be put any closer than they are in Fig. 1.4a. In building a three-dimensional structure, another close packed plane can be placed over the first, such that its spheres fit into the triangular cavities between the spheres of the first layer. There are two sets of such cavities, designated *B* and *C* in Fig. 1.4a. The second plane of atoms, Fig. 1.4b or c, can rest vertically over either the *B* positions or the *C* positions. If we assume that they rest over *B* positions, then the third close packed plane can be stacked on the second in two ways. The atoms of the third plane can be vertically over the *C* positions or alternatively over the first-plane atoms, i.e. over *A* positions. These two arrangements are designated *ABC* and *ABA* stackings respectively.

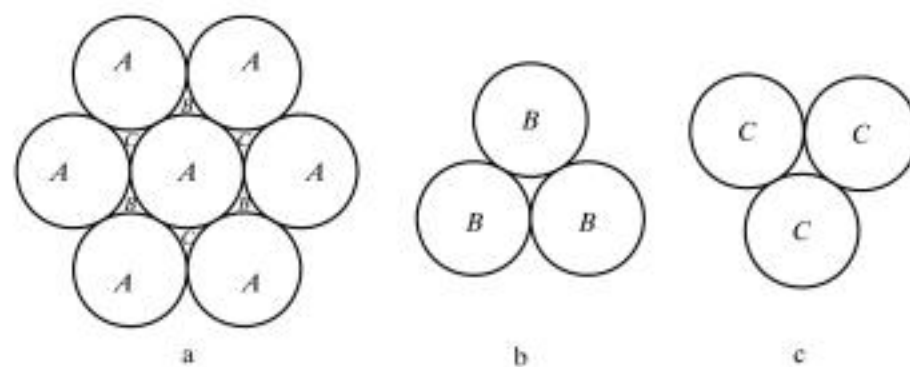


Figure 1.4 The geometry of close packing.

The FCC crystal is a stacking of close packed planes in the sequence ...*ABCABC*.... In Fig. 1.5, the correspondence between the ...*ABCABC*... stacking and the atoms of the unit cell is shown. This figure should be compared with Fig. 1.2b, where the atoms are drawn to their correct sizes. The close packed planes are inclined at an angle to the cube faces and are known as *octahedral planes*.

The HCP crystal is a stacking of close packed planes in the sequence ...*ABAB*.... Referring to Fig. 1.3b, if we call the close packed plane corresponding to the bottom hexagonal face *A*, it is easily seen that the atoms of the top hexagonal layer are vertically over those of the bottom layer and are,

therefore, to be designated also as *A*. The three atoms in the middle fit into the triangular cavities between the top and the bottom layers and are *B* plane atoms. The *C* positions are not used in this stacking.

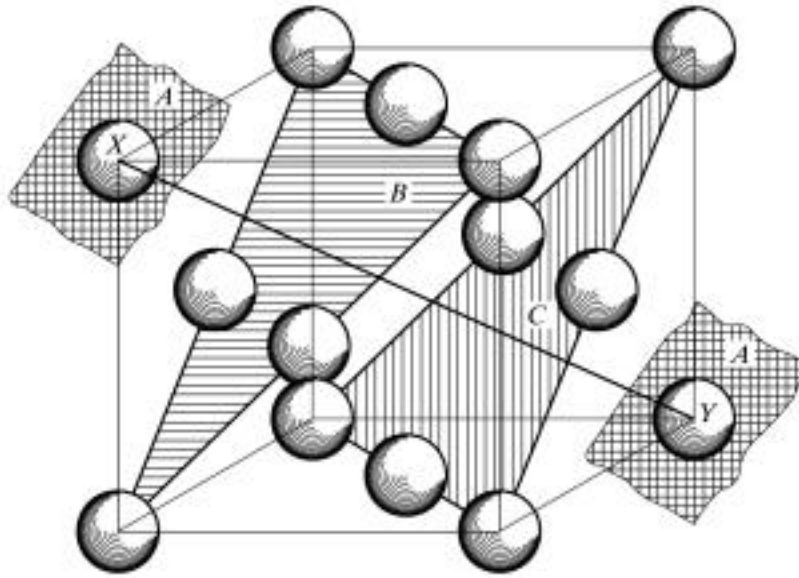


Figure 1.5 The correspondence between $\dots ABCA \dots$ stacking and the FCC unit cell. The body diagonal XY is perpendicular to the close packed planes.

The ideal c/a ratio of HCP stacking is 1.633. The ratios in real crystals, however, vary significantly from the ideal value, see table on page 5.

Miller indices The method devised by Miller is widely used to denote directions and planes in crystals. Let the vector $\mathbf{r}$ in Fig. 1.6a represent a direction in a crystal (not cubic in this case). To find the Miller indices of this direction, the components of the vector along the three axes are determined as multiples of the unit vector corresponding to each direction.

The vector $\mathbf{r}$ has a component of 2 along the $\mathbf{a}$ -axis, 1 along $\mathbf{b}$ and 1 along $\mathbf{c}$. Note that the absolute magnitudes of the components along $\mathbf{b}$ and $\mathbf{c}$ are not the same, but they are unity, when expressed as multiples of the corresponding unit vectors. The Miller indices of this direction is given by $[211]$, where the square brackets stand for a direction. If there is a negative component, it is denoted by a bar placed on top of the appropriate number.

In cubic crystals, it is convenient to have a notation to specify a *family of directions*. The family is denoted by pointed brackets, e.g. $\langle 211 \rangle$ stands for $[211]$, $[2\bar{1}1]$, $[121]$, $[11\bar{2}]$, $[12\bar{1}]$, etc. giving all combinations of the indices both positive and negative. The cube edges in the unit cell are represented by $\langle 100 \rangle$, the face diagonals by $\langle 110 \rangle$ and the body diagonals by $\langle 111 \rangle$.

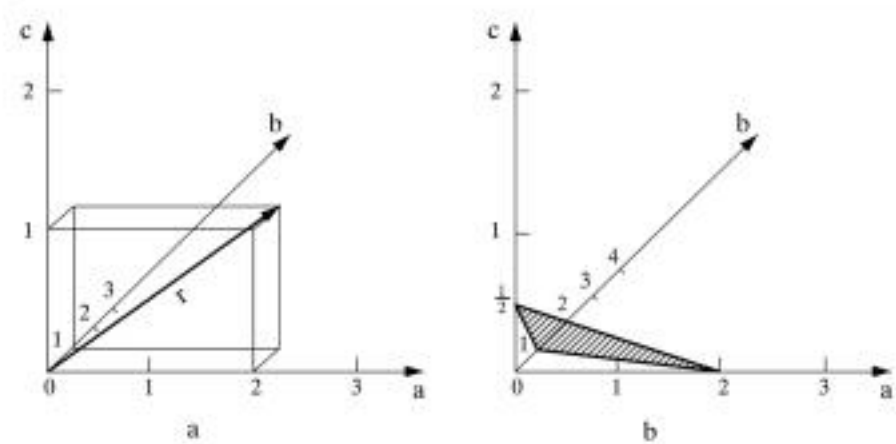


Figure 1.6 The Miller indices of (a) a crystal direction and (b) a crystal plane.

For determining the Miller indices of a plane in a crystal, the following procedure is used. Referring to Fig. 1.6b, the steps are :

1. Find the *intercepts* of the plane on the three coordinate axes
2. Take reciprocals
3. Convert into smallest integers in the same ratio
4. Enclose in parentheses

2	1	1/2
1/2	1	2
1	2	4
(124)		

A family of planes is denoted by curly brackets { }. For example, the faces of a cube are denoted by {100}. The octahedral planes in Fig. 1.5 are {111}.

- Example**
1. Draw a close packed plane within the unit cell of an FCC crystal and give its Miller indices.
 2. Mark the close packed directions that lie on this plane and give their Miller indices.
 3. Draw a (110) plane within the same unit cell and find the line of intersection of the two planes.

Solution

1. Referring to Fig. 1.5, the close packed planes in the FCC crystal are of the {111} type. Using D as the origin in Fig. 1.7, the Miller indices of the plane sketched is $(\bar{1}\bar{1}1)$.
2. The atoms are touching one another along the face diagonals of an FCC crystal. There are three face diagonals that lie on the above plane, corresponding to the three close packed directions. Their Miller indices are :
Taking origin at A, $\mathbf{AB} = [011]$
 at B, $\mathbf{BC} = [\bar{1}0\bar{1}]$
 at C, $\mathbf{CA} = [1\bar{1}0]$

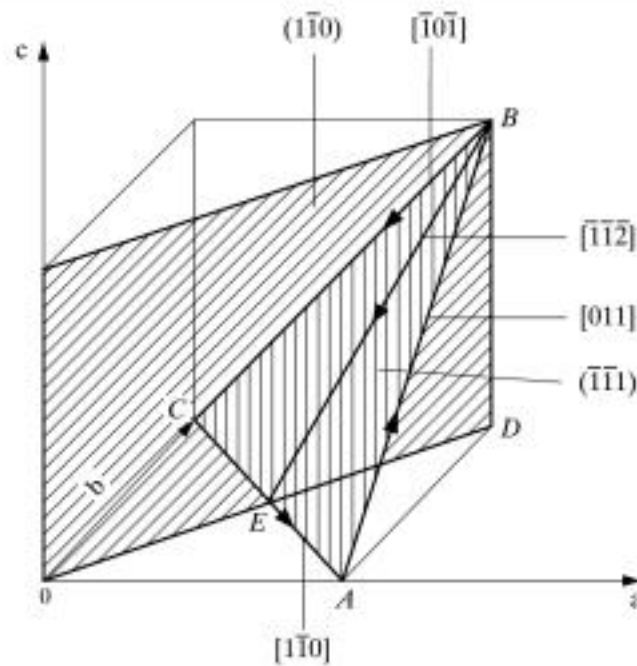


Figure 1.7 Planes and directions within the cubic unit cell.

3. Taking C as the origin, the $(1\bar{1}0)$ plane is drawn. The line of intersection of the two planes is BE .

Taking origin at B , $\mathbf{BE} = [\bar{1}\bar{1}\bar{2}]$

Note: As the choice of origin is arbitrary, it is chosen such that the directions start from the origin and the planes make intercepts on the three axes within the unit cell.

Miller-Bravais indices For hexagonal crystals, a four-digit notation $hkil$, known as Miller-Bravais indices, is used. The use of such a notation enables crystallographically equivalent planes or directions in a hexagonal crystal to be denoted by the same set of indices. Three of the axes $\mathbf{a}_1$, $\mathbf{a}_2$ and $\mathbf{a}_3$ are coplanar and lie on the basal plane of the hexagonal prism, Fig. 1.3, with a 120° angle between them. The fourth axis is the $\mathbf{c}$ axis perpendicular to the basal plane.

The determination of the Miller-Bravais indices is illustrated in Fig. 1.8 for the two examples given below. The indices of a prismatic plane (one of the vertical faces of the prism) is of the type $(10\bar{1}0)$. It makes an intercept of 1 and -1 along $\mathbf{a}_1$ and $\mathbf{a}_3$ axes respectively and is parallel to the $\mathbf{a}_2$ and $\mathbf{c}$ axes. Note that the use of three coplanar vectors results in the condition that $h + k = -i$.

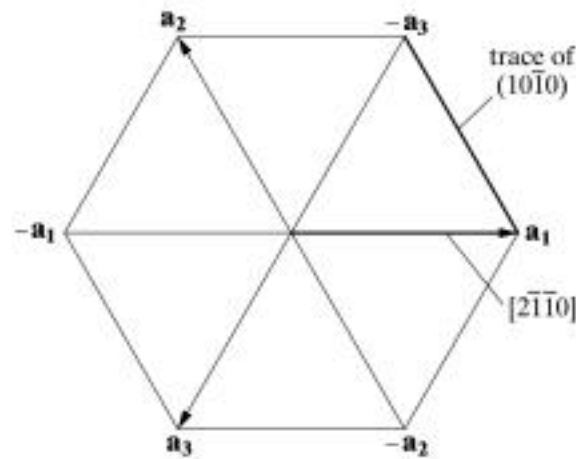


Figure 1.8 The Miller-Bravais indices for hexagonal crystals.

The indices of a direction that lie along one of the three $\mathbf{a}$ axes (parallel to a side of the hexagonal face) are of the type $[2\bar{1}\bar{1}0]$. For this example, the direction is parallel to the $\mathbf{a}_1$ axis and is resolved into components along $\mathbf{a}_2$ and $\mathbf{a}_3$. Each of these components being -1 , the value of the first index corresponding to the $\mathbf{a}_1$ axis can be obtained as $-(-1-1) = 2$ from the condition that $h + k = -i$. This yields $[2\bar{1}\bar{1}0]$.

1.3 STRUCTURE OF ALLOYS

Types of solid solutions When two metals are melted together and crystallized, a single crystal structure may form. In the unit cell of this crystal, both the metal atoms are present in proportion to their concentration. This structure is known as a *solid solution*.

Figure 1.9 illustrates three types of solid solutions. Figure 1.9a is a *random substitutional* solid solution. Here the solute atoms (shown shaded) have substituted randomly for the parent atoms (not shaded) on their sites. Figure 1.9b shows an *ordered* (substitutional) solid solution. Here, the solute and the solvent atoms are arranged in a regular fashion on the atomic sites. Perfect order is possible, only when the two metals are mixed in some fixed proportion like 1:1, 3:1, etc.

Figure 1.9c shows an *interstitial* solid solution. Here, the solute atoms are much smaller than the parent atoms and have occupied randomly the interstitial voids between parent atoms. In the FCC unit cell, one set of interstitial voids is located at the mid-points of the cube edges, $(0, 0, \frac{1}{2})$, $(0, \frac{1}{2}, 0)$, $(\frac{1}{2}, 0, 0)$ etc. and at the body centre $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$. These voids are called *octahedral voids*. The size of the atom that can fit into an octahedral void without distortion is $0.414r$, where r is the radius of the parent atoms that surround the void. Smaller

tetrahedral voids are located on the body diagonals at $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, $(\frac{3}{4}, \frac{3}{4}, \frac{3}{4})$ and other equivalent positions in the FCC unit cell. The size of the atom that can fit the tetrahedral void is $0.225r$. In the HCP crystal, the same types of octahedral and tetrahedral voids are present. In the BCC crystal, the tetrahedral voids at $(\frac{1}{2}, \frac{1}{4}, 0)$ positions are larger ($0.29r$), as compared to the octahedral voids at $(0, 0, \frac{1}{2})$ of size $0.15r$. Carbon atoms in BCC iron are located at the smaller $(0, 0, \frac{1}{2})$ voids.

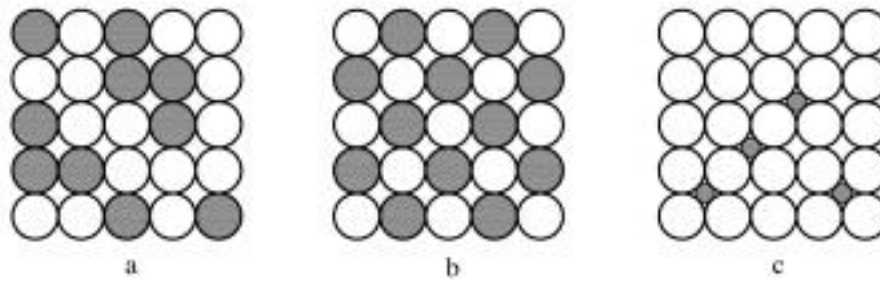


Figure 1.9 (a) Random substitutional solid solution, (b) ordered substitutional solid solution, and (c) interstitial solid solution.

Hume Rothery's rules Hume Rothery studied a number of alloy systems and formulated the conditions that favour *extensive substitutional solid solubility*. These are known as Hume Rothery rules and are listed below.

1. The size difference between the parent atom and the solute atom must be less than 15%.
2. The electronegativity difference between the metals must be small.
3. The crystal structure of the metals and the valency of the atoms must be the same.

The more is the size difference, the more intense is the stress field around the solute atom. This is valid for both large and small atoms. A large atom induces a compressive stress field around it, whereas a small atom induces a tensile stress field. The stress field increases the potential energy of the crystal. So, *the solid solubility becomes more limited, as the size difference increases*.

The electronegativity difference between two elements can be computed from Pauling's electronegativity numbers. This difference is usually small between typical metallic elements.

Complete solid solubility, i.e. the mixing of the two atoms in all proportions can evidently occur only when the crystal structures are identical. In the unit cell of one of the metals, the atoms can be continuously substituted by the other, till the pure crystal of the substituting atom is obtained.

The valency must be the same for extensive solid solubility. When the valencies are different, it is generally found that an excess of electrons is readily tolerated rather than a deficiency of bonding electrons. For example, zinc with

two valence electrons dissolves up to 38% in copper, whereas copper with one valence electron dissolves only about 3% in zinc.

Free energy of solid solutions When two metals A and B are mixed to form a solid solution, the Gibbs free energy G changes. The change in free energy (the free energy of the solid solution minus the free energies of the two pure metals) is called the free energy of mixing ΔG_{mix} . It can be written in terms of the enthalpy of mixing ΔH_{mix} and the entropy of mixing ΔS_{mix} :

$$\Delta G_{\text{mix}} = \Delta H_{\text{mix}} - T\Delta S_{\text{mix}} \quad (1.1)$$

1. For an *ideal solution*, $\Delta H_{\text{mix}} = 0$. Here, the A-A, B-B and A-B bonds are all *equally* favoured.
2. If $\Delta H_{\text{mix}} > 0$ (positive deviation from ideality), the formation of like bonds (A-A and B-B bonds) is preferred. For small positive values of ΔH_{mix} , there is a tendency for like atoms to *cluster* together in the solid solution. On an average, an A atom has more A neighbours than is the case for a perfectly random solid solution. At low temperatures or for large positive values of ΔH_{mix} , the clustering tendency increases. Eventually, *immiscibility* results. The solid solution splits into two separate phases, one rich in A and the other rich in B.
3. If $\Delta H_{\text{mix}} < 0$ (negative deviation from ideality), the formation of unlike bonds (A-B bonds) is preferred. Here, there is a tendency for *short range ordering*, i.e. A and B atoms show a preference to become nearest neighbours. At low temperatures, long range order may set in and an ordered solid solution may form. If ΔH_{mix} is only slightly negative, the ordering tendency may be so weak that long range ordering does not occur at any temperature. If ΔH_{mix} has a large negative value, the alloy may remain in the fully ordered state right up to the melting point, behaving like a chemical compound.

On mixing two metals at a constant temperature, a large number of distinguishable configurations w of equal potential energy result. The configurational entropy S_c is given by

$$S_c = k \ln w \quad (1.2)$$

where k is Boltzmann's constant. The entropy of mixing X_A mole fraction of A with X_B mole fraction of B ($X_A + X_B = 1$) in a random solid solution turns out to be

$$\Delta S_{\text{mix}} = -R (X_A \ln X_A + X_B \ln X_B) \quad (1.3)$$

X_A and X_B being fractions, the entropy of mixing is always positive. It tends to lower the free energy according to Eq. (1.1).

An ideal solid solution is random. For small deviations from ideality (either positive or negative), the solid solution can still be taken to be random for the purpose of computing the entropy of mixing. This is called a *regular solution* approximation. The free energy of mixing is then given by

$$\Delta G_{\text{mix}} = \Delta H_{\text{mix}} + RT(X_A \ln X_A + X_B \ln X_B) \quad (1.4)$$

The variation in the free energy of a solid solution as a function of composition is sketched schematically in Fig. 1.10. Figure 1.10a is for an ideal solution, where $\Delta H_{\text{mix}} = 0$. The free energy decreases with increasing concentration of either element reaching a minimum at the mid-composition, where the entropy of mixing is a maximum. Figure 1.10b is a case where $\Delta H_{\text{mix}} < 0$. The free energy curve has the same shape as in Fig. 1.10a, but more U-shaped. Figure 1.10c is for a solution where $\Delta H_{\text{mix}} > 0$. The free energy curve decreases initially at the two pure ends due to the dominance of the entropy term. With increasing concentration, however, the positive ΔH_{mix} term dominates and the free energy curve has two minima and a maximum. There is no complete solid solubility in this case. The maximum solid solubilities are obtained by drawing the common tangent xy to the free energy curve, Fig. 1.10c. The compositions corresponding to the points of tangency c_1 and c_2 give the limit of solubility at the two ends respectively. In between c_1 and c_2 , the alloy is stable only as a mixture of two solid solutions of composition c_1 and c_2 . With increasing temperature, however, the entropy term may dominate over the enthalpy term for all compositions, the free energy curve may become U-shaped and a single solid solution may become stable over the entire composition range.

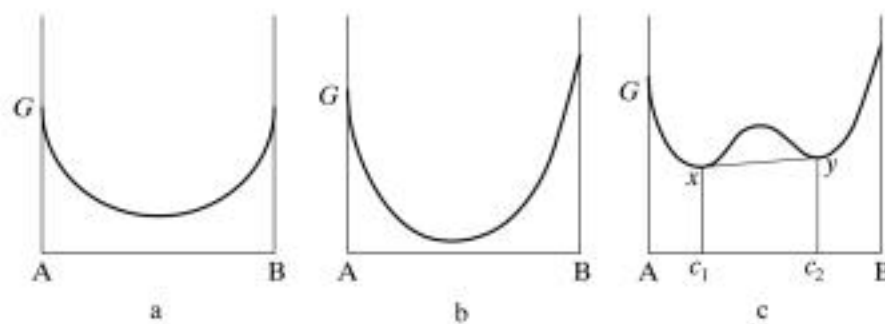


Figure 1.10 Free energy of solid solutions: (a) enthalpy of mixing ΔH_{mix} is zero (ideal solution), (b) ΔH_{mix} is negative, and (c) ΔH_{mix} is positive.

Intermediate phases Intermediate phases refer to alloy crystal structures which are different from either of the parent crystals. Some intermediate phases have a fixed composition and are called *intermetallic compounds*. Quite often, however, the intermediate phases have a significant composition range.

Only a few intermetallic compounds obey the *normal rules of valency*. Many of these are based on the cubic structures of sodium chloride (NaCl) or calcium fluoride (CaF_2). In the NaCl structure, the chlorine ions are at the FCC positions of the cubic unit cell and the sodium ions are at the octahedral voids. Examples of such intermediate phases are MgSe, PbSe and CdTe. In the CaF_2 structure, the calcium ions are at the FCC positions and the fluorine ions are at the tetrahedral void positions. Examples are Mg_2Si , Mg_2Sn , Cu_2S and Be_2C .

The *nickel arsenide* (NiAs) structure is often formed by the transition metals alloyed with the metalloids of the V and VI columns such as As, Bi, Sb, Se and

Te. The structure has the HCP stacking sequence with alternate layers of metal and metalloid atoms. These phases have a fairly wide range of solubility. Examples are CrSb, CuSn and MnSe.

Electron phases (also called electron compounds or Hume Rothery compounds) have a wide range of solubility. They occur at certain definite values of valence electrons to atom ratio in the alloy such as 3:2, 21:13 and 7:4. Typical examples are CuZn (3:2), Cu₅Zn₈ (21:13) and CuZn₃ (7:4).

Laves phases possess one of the three related structures with the general formula AB₂. Examples of the three prototype structures based on magnesium are: MgCu₂ (cubic), MgZn₂ (hexagonal) and MgNi₂ (hexagonal).

The *sigma phase* has a complex crystal structure and is very brittle. The unit cell is tetragonal with 30 atoms in the cell. This phase is a source of embrittlement in stainless steels.

Alloy carbides and nitrides form very hard and brittle compounds. They are often cubic with the formula MX, with the metal atoms (M) in FCC positions and the carbon or nitrogen (X) in interstitial positions (octahedral or tetrahedral). Examples are TiC, VC, NbC, TiN, VN and NbN. Examples of carbides with other structures are Cr₂₃C₆, Cr₇C₃, MoC and WC.

Like other solid solutions, intermediate phases may remain disordered at high temperatures and become ordered at low temperatures. The β (BCC) phase of the Cu-Zn alloy system is a typical example. The composition of β is around 50Cu50Zn. At temperatures above 465°C, the copper and zinc atoms are randomly distributed on the BCC sites. At lower temperatures, an ordered structure forms with the zinc atoms at the body corners and the copper atoms at the body centres (or vice versa). When this occurs, extra lines appear in the diffraction pattern, as the ordered structure corresponds to a simple cubic space lattice (with lattice points at the eight corners of the cube only) rather than the BCC space lattice. The ordered compound is known as a *superlattice*, as it gives rise to additional diffraction lines.

In general, intermetallic compounds and ordered solid solutions possess little ductility and are relatively brittle phases. They are useful for increasing the strength of a ductile matrix, but at the cost of reduced ductility.

1.4 IMPERFECTIONS IN CRYSTALS

Imperfections in crystals can be classified according to their geometry as point, line and surface imperfections. The characteristics of different imperfections are briefly considered below.

Point imperfections There are point-like regions of distortion in the crystal. The common types are :

- (a) vacancy
- (b) interstitialcy

- (c) interstitial impurity, and
(d) substitutional impurity

A *vacancy* is an atomic site from which the atom is missing, *a* in Fig. 1.11. An *interstitialcy* refers to an atom displaced from a regular site into an interstitial site, *b* in Fig. 1.11. An *interstitial impurity* is a small atom occupying an interstitial void space between parent atoms of the crystal, *c* in Fig. 1.11. A *substitutional impurity* is an impurity atom that has substituted for one of the parent atoms on a regular atomic site, *d* in the figure.

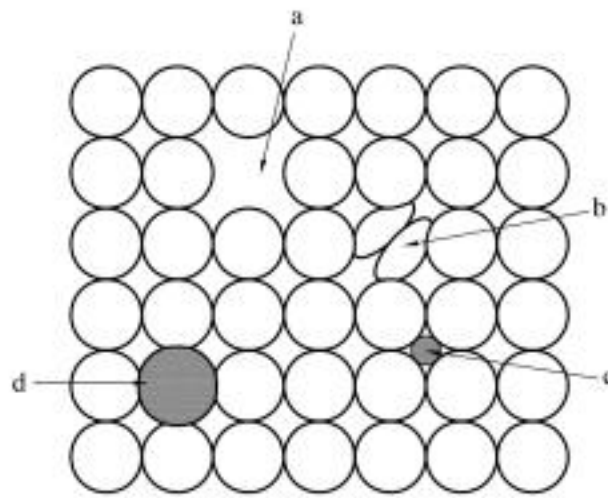


Figure 1.11 Point imperfections in crystals: (a) vacancy, (b) interstitialcy, (c) interstitial impurity, and (d) substitutional impurity.

Point imperfections increase significantly the configurational entropy of the crystal and thereby lower its free energy. So, a certain number n of point imperfections is in thermodynamic equilibrium in the crystal at a given temperature. For example, the fraction of vacancies n/N in a crystal at temperature T is given by

$$\frac{n}{N} = \exp\left(-\frac{\Delta H_f}{kT}\right) \quad (1.5)$$

where N is the total number of atomic sites and ΔH_f is the enthalpy of formation of a vacancy and is equal to the reversible work done at constant pressure in bringing an atom from inside the crystal to the surface.

Dislocations Line imperfections are known as dislocations. The geometry of dislocations can be understood with reference to the two limiting straight line types:

1. the edge dislocation, and
2. the screw dislocation



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glides on a (111) plane causing slip over the hatched region. It then cross-slips on to a nonparallel $(\bar{1}\bar{1}1)$ plane. In the double cross-slip process illustrated, it again cross-slips back to a (111) plane, which is parallel to the initial plane.

Edge dislocations cannot cross-slip. When they move on a plane other than the slip plane, the motion involves *addition or subtraction of atoms to the edge*. This motion is nonconservative and is called *climb motion*. Addition or subtraction of atoms generally occurs at an atomic step called *jog* on the edge of the incomplete plane, Fig. 1.14. Edge dislocations climb up or climb down, depending on whether the incomplete plane shrinks or increases in extent. Addition of atoms is required for climbing down, whereas subtraction of atoms (or equivalently addition of vacancies) is required for climbing up. In Fig. 1.14, moving of the jog to the left is equivalent to climbing down; movement to the right is climbing up. Climb can clearly occur only when there is sufficient thermal energy for the diffusion of atoms to or away from the edge of the incomplete plane. It is, therefore, an important process in high temperature deformation known as *creep*, discussed under Sec. 5.4.

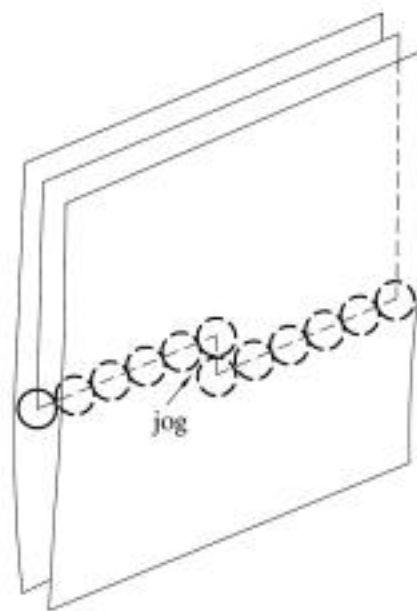


Figure 1.14 Jog is a step on the edge of the incomplete plane of an edge dislocation.

Partial dislocations and stacking faults Full dislocations described above have an integral lattice translation as the Burgers vector. For partial (or imperfect) dislocations, the Burgers vector is a fraction of an interatomic distance. As the elastic energy of dislocations is proportional to the square of their Burgers vector, the splitting of a full dislocation b_1 into two partial dislocations b_2 and b_3 is favoured energetically, when



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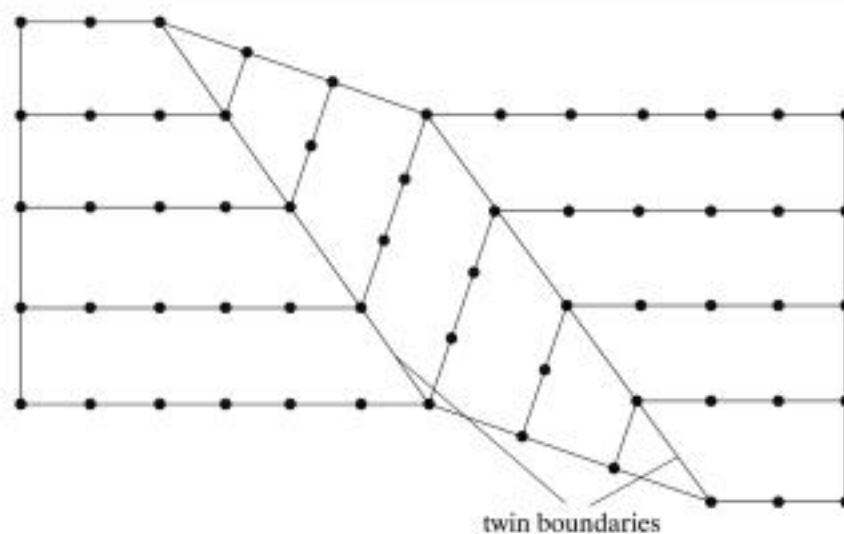


Figure 1.17 The atomic arrangement on one side of a twin boundary is a mirror reflection of that on the other side.

A large substitutional atom can interact with a vacancy in the same crystal to lower the distortional energy. The two defects stay together in close proximity. Point imperfections interact with the distortional geometry associated with dislocations. For example, a large substitutional atom can lower its distortional energy by staying in the tensile region just below the edge of the incomplete plane of an edge dislocation. An interstitial atom, that is too large for the void it occupies, can similarly lower its energy by staying in the core of an edge dislocation. Such an effect is particularly marked in low carbon and mild steels, where the oversized (interstitial) carbon atoms tend to segregate in voids around the tensile region of edge dislocations. Such solute atmospheres around dislocations are known as *Cottrell atmospheres*. Point imperfections also segregate at grain boundaries as the structure at the boundary is more open than that in the interior of the crystal. Such segregation results in several metallurgically important effects such as the solute drag effect on a migrating grain boundary and intergranular cracking.

Dislocations lie on slip planes within the crystal. As the crystal orientation across a grain boundary (or a twin boundary) changes abruptly, a dislocation on a slip plane in any one grain cannot cross the boundary and move on to the same slip plane of the next grain. Because of this, moving dislocations get piled up against grain boundaries. This results in an important influence of grain size on the yield strength of metals and alloys.

1.5 EXPERIMENTAL STUDY OF STRUCTURE

The structure seen with the naked eye or under low magnification by acid etching of steel bars, billets, castings and forgings is called *macrostructure*. It



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electropolishing. In the window technique, pointed electrodes are used. The metal directly in front of the pointed electrode is dissolved more rapidly and perforates. The metal adjacent to the perforation is thin enough to transmit the electrons. Diffraction patterns can be obtained over any selected area of the matrix or of particles embedded in it.

The scanning electron microscope In the *scanning electron microscope*, Fig. 1.19b, a beam of electrons is moved over the specimen surface in a sequence of closely spaced lines. The electrons interact with the surface atoms and the output depends on the composition and topography. The output signal is picked up and used to modulate the brightness of a cathode ray beam, which is moving synchronously with the scanning search beam. The depth of focus of the scanning electron microscope is so high that a fracture surface can be directly examined without any polishing. In metallurgical studies, fractographic examination and, to a lesser extent, examination of polished and etched surfaces are carried out with the scanning electron microscope.

The electron probe microanalyzer The composition of alloys can be determined by conventional chemical analysis and other modern instrumental methods such as spectroscopy or x-ray fluorescence. These give only the average composition. The electron probe microanalyzer (EPMA) is a tool for determining the composition on a micron scale relevant to metallurgical problems such as identification of very small inclusions and precipitates and composition gradients on a micron scale. In this instrument, a beam of electrons about $\frac{1}{2}$ μm diameter strikes the specimen surface, interacting with the atoms of the specimen to produce x-rays. By measuring the wavelength and intensity of the x-rays produced, it is possible to determine which elements are present and their respective concentrations.

X-ray diffraction The crystal structure of metals and alloys is determined by x-ray diffraction. The Bragg law describes the effect produced by a beam of x-rays of wavelength λ incident at an angle of θ (called the Bragg angle) on to a set of parallel atomic planes with an interplanar spacing of d :

$$n\lambda = 2d \sin \theta \quad (1.8)$$

where n is an integer.

There are a number of methods for determining the crystal structure. In the powder method, a monochromatic beam of x-rays is incident on the powder sample of the unknown metal. The diffracted beam is recorded in a diffractometer as a function of 2θ . Figure 1.20 shows a tracing from a diffractometer, where the intensity peaks correspond to positions, where the Bragg condition given in Eq. (1.8) is satisfied. Computer software packages are now available for analyzing the diffraction data and ascertaining the crystal structure.



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Components Components refer to the *independent chemical species* that constitute an alloy. For plain carbon steel, iron and carbon are the chemical species; so, the components are Fe and C (graphite). However, carbon almost always occurs in the combined form as cementite (Fe_3C) in steels, even though cementite is metastable and has a higher free energy as compared to graphite. So, it is common to show the phase relationships in the Fe-C system on a metastable phase diagram with Fe and Fe_3C as the components. The choice of all the three (Fe, C and Fe_3C) as components does not constitute a set of independent chemical species.

Degrees of freedom The degrees of freedom refer to the number of *independent variables* associated with the system. The total number of variables is the external variables (temperature and pressure) plus the composition variables. The number of composition variables required to express the concentration of C components in a phase is $(C - 1)^*$. For a system with P phases,

$$\text{total variables} = P(C - 1) + 2 \quad (2.2)$$

The degree of freedom F is equal to or less than the total number of variables :

$$F = C - P + 2 \leq P(C - 1) + 2 \quad (2.3)$$

When there is only one phase in a system, F is equal to the total number of variables. For $P > 1$, F is less than the total number. The minimum value for the degree of freedom is zero and this sets an upper limit on the number of phases that can coexist in a system under equilibrium.

2.2 BINARY PHASE DIAGRAMS

The most commonly used phase diagrams are those that depict the equilibrium between two components. These are known as *binary phase diagrams*. Many engineering alloys contain more than two components. Nevertheless, the binary diagram between the two major components is still useful in interpreting the phase relationships. For example, the Fe- Fe_3C phase diagram is used in the study of carbon steels, which usually contain small quantities of S, P, Si and Mn, in addition to Fe and C. In other cases, more complex phase relationships have to be studied. For equilibrium in a system containing three components, ternary phase diagrams are used. Further, it is possible to depict the more complicated phase relationships between a number of components on a pseudo-binary diagram.

For graphical representation of the binary phase relationships, we need a map with three axes corresponding to temperature, pressure and composition. As three-dimensional diagrams are difficult to draw on paper, for binary phase diagrams it is customary to ignore the pressure variable and the vapour phase.

*Note that every phase in equilibrium in a system will have all the components in it, however small may be the concentration of a component.



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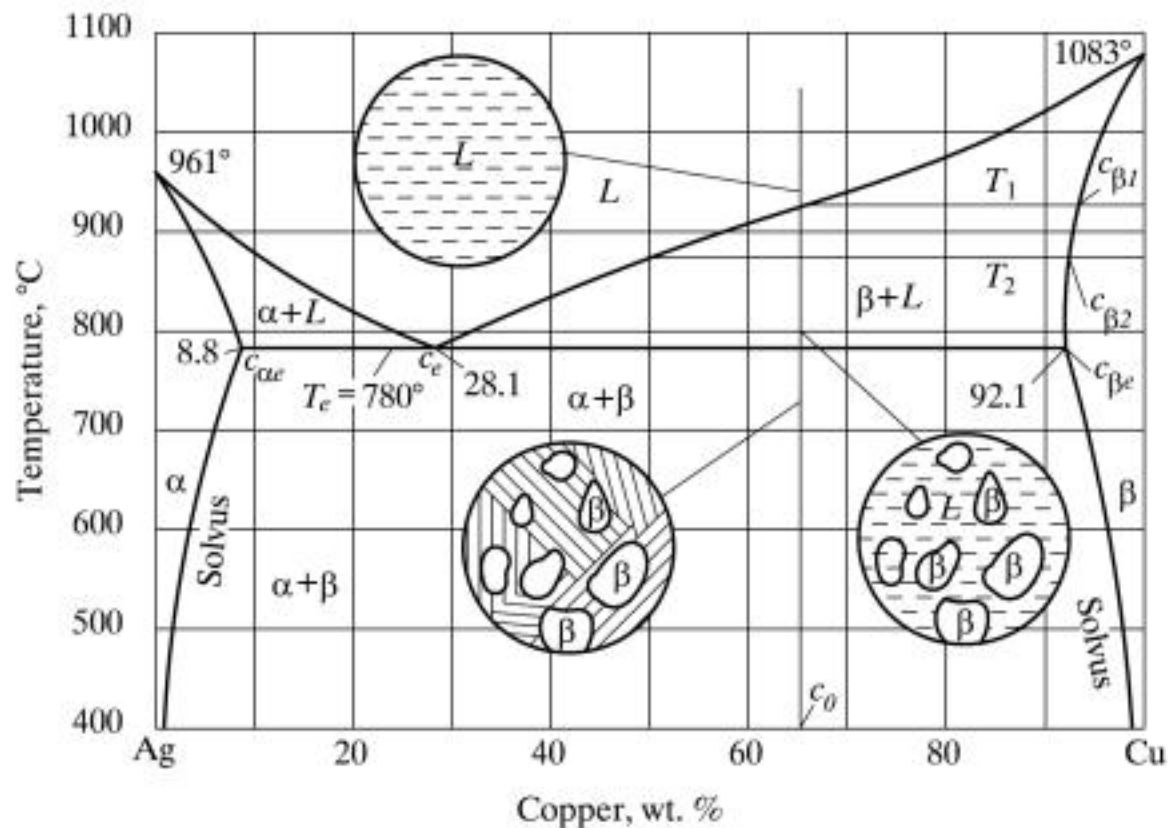


Figure 2.2 The silver-copper phase diagram is a simple eutectic system.

reaction	:	L	$\xrightleftharpoons{\text{cool}}$	α	+	β
structure	:	—		FCC		FCC
composition	:	c_e		$c_{\alpha e}$		$c_{\beta e}$
wt% Cu	:	28.1		8.8		92.1

Plate-like crystals (or rod-like crystals) of α and β arranged in an alternating, parallel fashion are the characteristic structure of the eutectic mixture as shown in Fig. 2.3a. Sometimes, an *abnormal* or *divorced* eutectic structure is observed. Here, the eutectic is present in so small a volume fraction that the plate-like structure does not form. Instead the minor phase of the eutectic forms and grows as islands that appear unconnected on a polished surface. The other phase forms the continuous matrix. Abnormal eutectic structures are observed, when one of the phases forms strong covalent bonds, e.g. the Al-Si and Fe-graphite systems. Chinese script is a structure between the lamellar (normal) eutectic and the abnormal eutectic. Here, the lamellar morphology has degenerated into a pattern that looks like a Chinese script, Fig. 2.3b.

An alloy to the left of the eutectic is known as a *hypoeutectic* alloy. An alloy to the right is called a *hypereutectic* alloy. In both these cases, on cooling below the liquidus boundary, the *proeutectic* phase (α or β) separates out first. On further cooling, the fraction of the proeutectic phase increases, till the eutectic temperature is reached. Just above the eutectic temperature T_e , the tie-line indicates that the composition of the proeutectic phase is $c_{\alpha e}$ (or $c_{\beta e}$) and the liquid is of



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lower temperature T_2 cuts across the two phase region in Fig. 2.5a. The corresponding free energy curves intersect each other, Fig. 2.5c. There is a certain range of composition over which the free energy of a mixture of the liquid and the solid phases is lower than either of the phases. This composition range is obtained by drawing a *common tangent* xy to the two curves, see Fig. 2.5c. The compositions at points x and y are c_l and c_s respectively. Over the range of composition to the left of c_l , the free energy of the liquid is lower; so, it is the stable phase. For compositions that lie between x and y , two-phase equilibrium prevails. The mixture of the two phases in the proportion given by the lever rule has the lowest free energy. For compositions to the right of c_s , the solid phase is the stable phase. At temperature T_3 , Fig. 2.5d, the solid phase has lower free energy and is stable over the entire composition range.

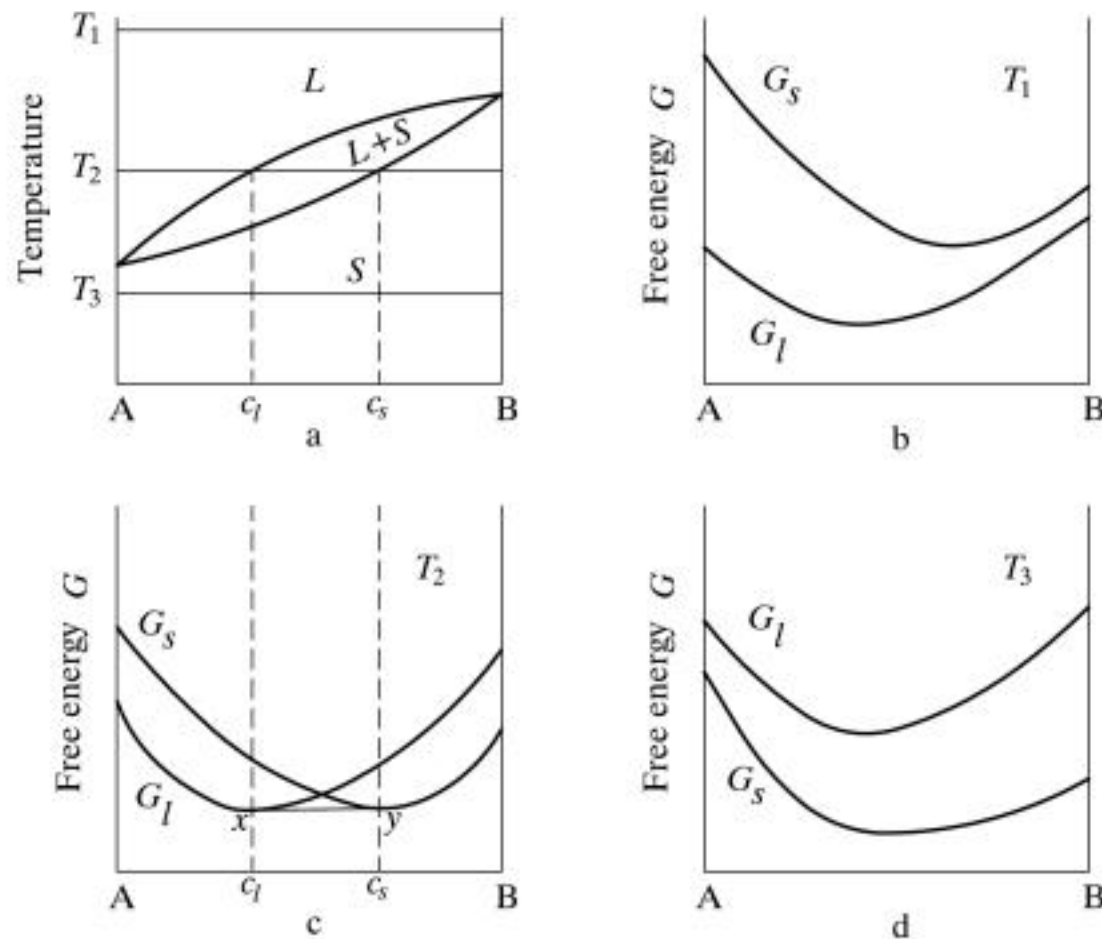


Figure 2.5 Free energy-composition relationships for an isomorphous system at three temperatures.

For a eutectic system, three free energy-composition curves are drawn, one for each of the phases: α , β and liquid. The sketches in Fig. 2.6a-d illustrate the relationships. At T_1 , Fig. 2.6a, the liquid phase is stable over the entire composition range, as it has a lower free energy. At T_2 , the liquid free energy curve intersects both the α and β free energy curves, Fig. 2.6b. Two common



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Deviation from equilibrium is a serious problem in alloy systems in which a peritectic reaction occurs. Recall that this reaction is between a liquid and a solid. The product of the reaction is a solid, which forms at the boundary between the two parent phases. As soon as some solid product (α) forms, the parent phases (L and β) get separated and are no longer in contact. This phenomenon is known as *surrounding or enveloping*, Fig. 2.9. Further reaction

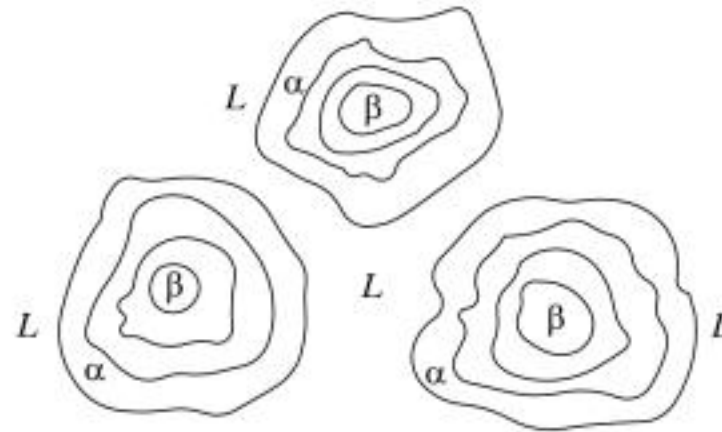


Figure 2.9 In the peritectic reaction $L + \beta \rightarrow \alpha$, surrounding or enveloping occurs, when the product α phase surrounds the unreacted β phase.

can occur only through solid state diffusion through the intervening product layer. Solid state diffusion being very slow, peritectic alloys solidify with large deviations from the equilibrium composition and structure.

A homogenizing treatment is required to remove the compositional variations that accompany coring or surrounding. The alloy is heated to a high temperature and held there for a prolonged period, during which diffusion levels out the compositional gradients and yields a homogeneous solid solution. Extreme care is required during heating not to cross the solidus. Note that, for a cored structure, the effective solidus is lower than the equilibrium solidus. If this solidus is crossed, melting occurs near the grain boundaries. Liquation is often accompanied by grain boundary oxidation and the phenomenon is called *burning*. A burnt alloy is permanently damaged, as the oxidized grain boundaries cannot be restored to the initial unoxidized condition by means of a heat treatment or by mechanical working. *Overheating* refers to heating close to the solidus, but without causing liquation. Overheating impairs the properties of steels through loss of ductility and toughness. It is associated with the segregation of impurities and subsequent precipitation along grain boundaries. The effects of overheating are not permanent and can be removed by a subsequent heat treatment.

2.5 THE IRON CARBON EQUILIBRIUM DIAGRAM

In Fe-C alloys, carbon may be present either in the form of free carbon (graphite) or as iron carbide (Fe_3C). The most frequently occurring form in steels is Fe_3C , known as *cementite*. As already pointed out, *cementite is metastable with respect*



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As the two boundaries of a cementite plate are close together, they may not be resolved as separate lines. Cementite often appears as a single dark line. At higher magnifications or in a coarse distribution in pearlite, the cementite boundaries show up as separate lines.

Hypoeutectoid steels The composition of hypoeutectoid steels is less than 0.77% C. In addition to pearlite, the *proeutectoid ferrite* (pro α) is present, Fig. 2.12. The fraction of the pro α can be estimated from lever rule. For example, in mild steel of 0.2% C,

$$f_{\text{pro } \alpha} = \frac{0.77 - 0.2}{0.77 - 0.02} = 0.76$$

$$f_{\text{pearlite}} = 1 - 0.76 = 0.24$$

The proeutectoid ferrite forms at the austenite grain boundaries between A_3 and A_1 temperatures.

Hypereutectoid steels Here, the proeutectoid phase is cementite. It separates between A_{cm} and A_1 temperatures as a thin network along the austenite grain boundaries. In a 1.2% C steel, applying lever rule, we have

$$f_{\text{pro Fe}_3\text{C}} = \frac{1.2 - 0.77}{6.67 - 0.77} = 0.07$$

$$f_{\text{pearlite}} = 1 - 0.07 = 0.93$$

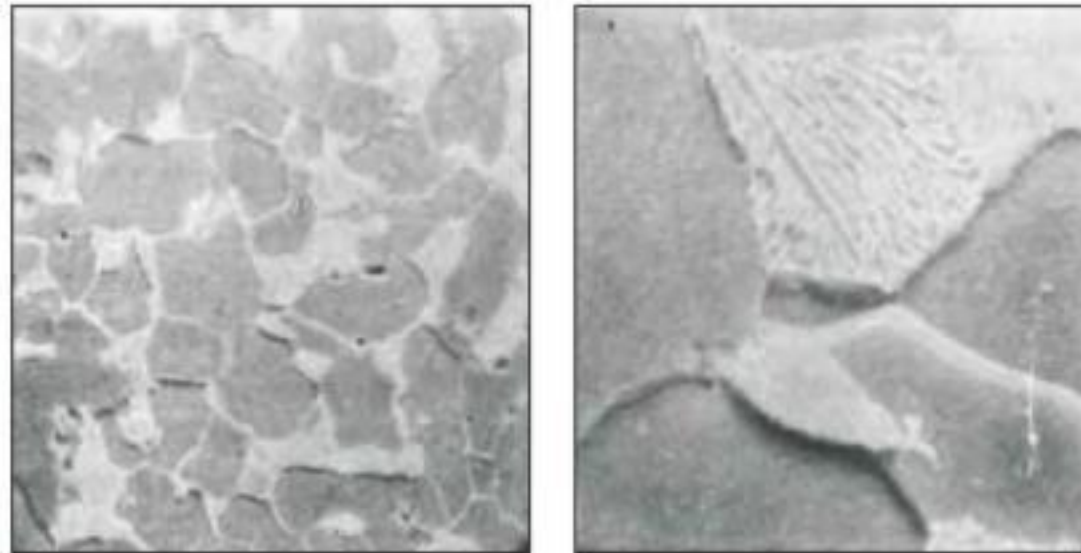


Figure 2.12 Scanning Electron Micrograph of mild steel, showing proeutectoid ferrite and pearlite.

2.6 EFFECT OF ALLOYING ELEMENTS ON THE Fe-C DIAGRAM

The effect of alloying elements on the Fe-C phase relationships is best understood by first considering the two types of phase diagrams obtained, when iron is alloyed with a substitutional element.



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move closer, thus restricting the extent of the austenite phase. To the left of the field are austenite and ferrite and to the right austenite and carbide. The eutectoid temperature moves up. The eutectoid composition shifts to lower carbon contents. With Cr > 19%, the austenite phase field disappears altogether. The critical composition at which this happens are given below for a few typical cases.

Ferrite stabilizer	Cr	Si	Mo	W	V
Critical composition, wt.%	19.5	8.1	6.4	10.8	4

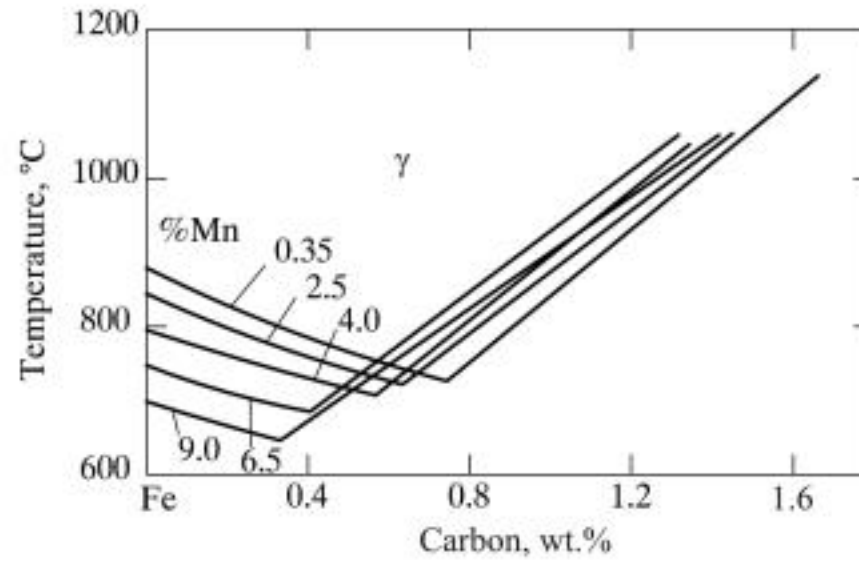


Figure 2.16 The shifting of the A_3 and A_{cm} phase boundaries in the presence of manganese.

Carbon being an austenite stabilizer, a higher concentration of a ferrite stabilizing element (19.5% Cr for Fe-C-Cr as compared to 12.7% Cr for Fe-Cr) is required in the presence of carbon to eliminate the austenite field.

For low alloy steels containing C between 0.08 and 1.4 wt%, A_{c1} and A_{c3} critical temperatures are given by the following empirical equations:

$$A_{c1} (^{\circ}\text{C}) = 723 - 10.7\text{Mn} - 16.7\text{Ni} + 29.1\text{Si} + 16.9\text{Cr} + 290\text{As} + 6.38\text{W}$$

$$A_{c3} (^{\circ}\text{C}) = 910 - 203\sqrt{\text{C}} - 30\text{Mn} - 20\text{Cu} - 15.2\text{Ni} - 11\text{Cr} - 700\text{P} + 44.7\text{Si} + 31.5\text{Mo} + 104\text{V} + 460\text{Al} + 13.1\text{W} + 120\text{As}$$



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$$f_{\alpha} = pr/qr$$

$$f_{\beta} = qp/qr$$

If three phases coexist, their relative proportion is determined from a tie-triangle. For example, if the overall composition o in Fig. 2.19b exists as a mixture of γ , δ and ϵ , the respective compositions being s , t and u , the relative proportions of the three phases are given by (see Fig. 2.19b):

$$f_{\gamma} = ol/sl$$

$$f_{\delta} = om/tm$$

$$f_{\epsilon} = on/un$$

An example of an isothermal section of the Fe-Cr-Ni system at 650°C is shown in Fig. 2.20. Two tie-triangles are seen in this section. Three phases are present in each of the tie-triangles.

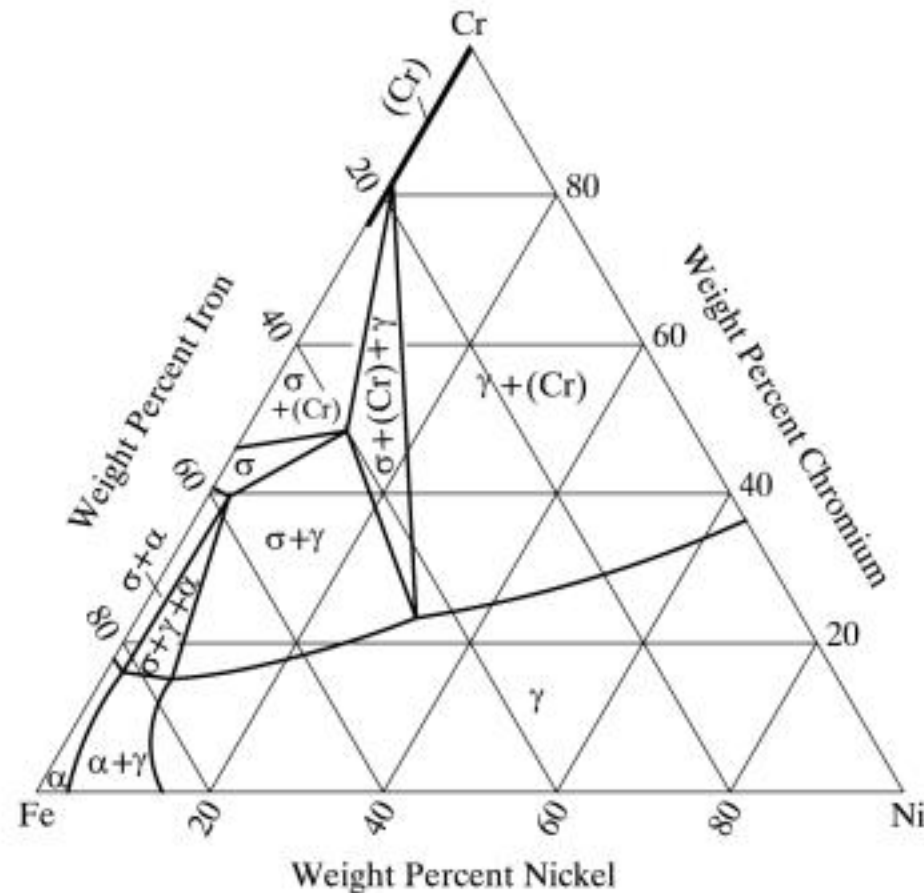


Figure 2.20 Fe-Cr-Ni ternary isothermal section at 650°C.

Another common way of presenting the ternary phase relationships is to draw vertical sections (called *isopleths*), keeping the concentration of one component constant. The x -axis then gives the variation of one of the other two components. Temperature is plotted on the y -axis. Figure 2.21 shows a vertical section of the Fe-Cr-Ni system at a constant Fe content of 74 wt%.



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For two-dimensional representation, the concentration of one of the components and temperature is to be kept constant. Figure 2.25 shows an isothermal section of the Fe-Cr-V-C system at 700°C and at a constant C content of 0.2 wt%. $M_{23}C_6$, M_7C_3 and MC are alloy carbides, with M standing for (Fe, Cr, V). ABC is the triangular section of a tie-tetrahedron, showing a four-phase equilibrium.

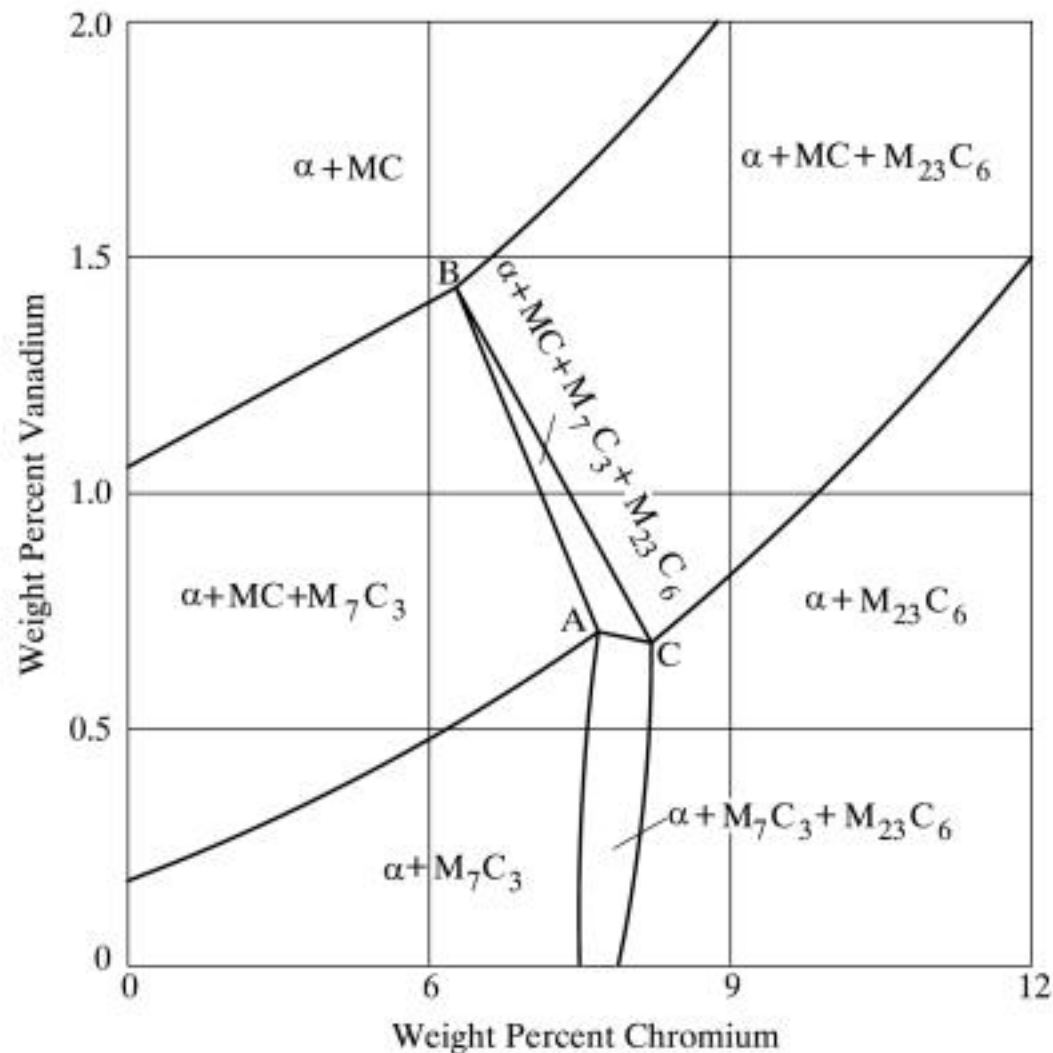


Figure 2.25 Fe-Cr-V-C quaternary isothermal section at 700°C and at a constant carbon content of 0.2 wt%.

2.10 EXPERIMENTAL DETERMINATION OF PHASE DIAGRAMS

A variety of physical techniques are used for the experimental determination of phase diagrams. Among these, the more important are:

- (i) thermal analysis,
- (ii) microscopic examination, and
- (iii) x-ray diffraction.

Other auxiliary techniques are dilatometry, electrical resistivity and magnetic measurements.



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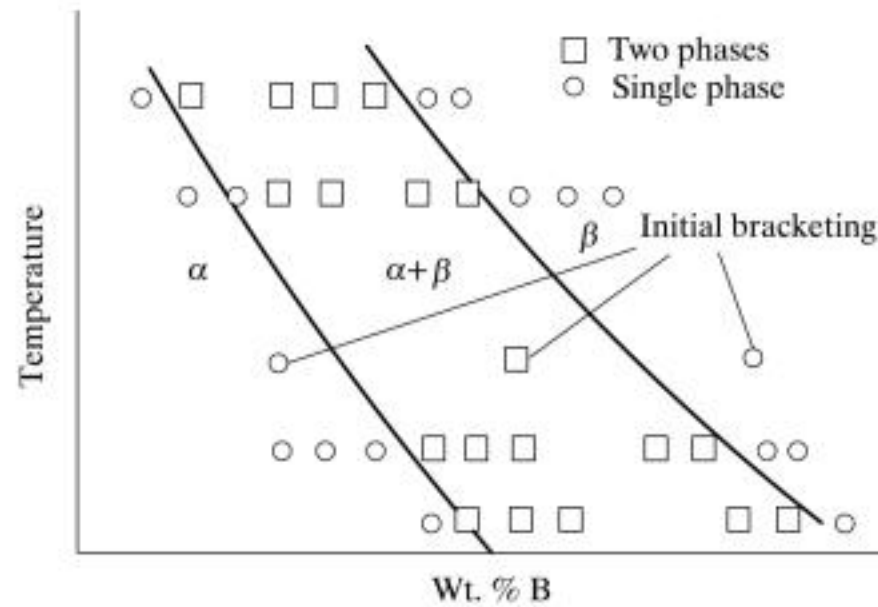


Figure 2.28 The determination of solvus boundaries by the metallographic and x-ray methods.

Auxiliary techniques *Dilatometry* is based on the volume change associated with most phase changes. It is common to record the length changes of a specimen as a function of temperature during heating or cooling. Figure 2.29

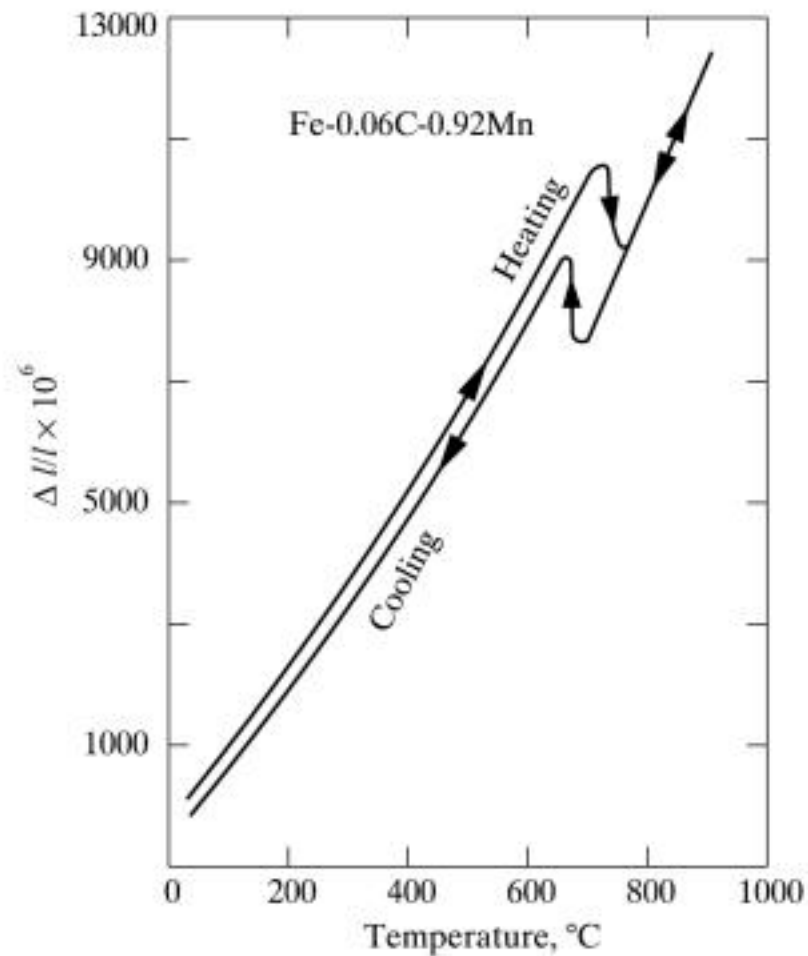


Figure 2.29 The fractional change in length of a low carbon steel on heating and cooling. The abrupt changes in slope correspond to phase changes.



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in intensity with time, until two discrete phases corresponding to the equilibrium compositions form. Many metallurgical transformations occur by nucleation and growth. We will consider only this mechanism.

3.2 DIFFUSION IN SOLIDS

Diffusion is the mass flow process in which atoms change their positions relative to their neighbours in a phase under the influence of thermal energy and a concentration gradient.

Fick's laws Fick's first law describes steady state diffusion and is written as

$$J = -D \frac{dc}{dx} \quad (3.1)$$

where J is the diffusional flux in the x -direction (number of atoms transported per unit area per unit time), dc/dx is the concentration gradient, and D is the diffusion coefficient.

Under steady-state conditions, the flux remains independent of time t and position x .

Fick's second law is applicable to nonsteady state diffusion. Here, the flux is changing as a function of position and time. When the diffusion coefficient is not a function of concentration, the second law is of the form:

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} \quad (3.2)$$

Solutions to the second law for different initial and boundary conditions are available, but will not be covered here. A quick but approximate estimate of the depth of penetration x during nonsteady state diffusion is given by

$$x = \sqrt{Dt} \quad (3.3)$$

Atomic model of diffusion In interstitial diffusion, Fig. 3.1a, the diffusing atom jumps from one interstitial site to a neighbouring interstitial site in executing a unit step. The neighbouring site is usually vacant in dilute solutions. The activation energy for interstitial diffusion is simply the energy barrier along the path from one interstitial site to the next. Substitutional diffusion occurs with the aid of vacancies, Fig. 3.1b. A substitutional atom jumps from a regular site into a neighbouring vacant site in executing the unit step. Here, in addition to considering the probability of an atom crossing the activation barrier along the path, the probability of the neighbouring site being vacant has also to be taken into account. Because of this, substitutional diffusion is generally 4–6 orders of magnitude slower than interstitial diffusion.

The unit step in diffusion is executed with the help of thermal energy. So, the diffusion coefficient D is strongly dependent on temperature and is of the form



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When both the parent and the product phases are solids, volume or shape changes introduce elastic strains during nucleation. The strain energy is added as a positive energy term to the nucleation Eq. (3.5). This has an additional inhibiting effect on the transformation. To minimize the total energy, the particle usually assumes a nonspherical shape such as a disc or a plate.

The increase in the size of the particle beyond the critical size is called *growth*. Like nucleation, growth is dependent on thermal fluctuations. The growth rate U is zero at the equilibrium temperature T_0 , as well as at 0 K. It passes through a maximum at some intermediate temperature, which turns out to be higher than that for the nucleation maximum, as shown in Fig. 3.4.

Heterogeneous nucleation Here, there are certain preferred sites in the parent phase, where the probability of nucleation is much greater than that at other sites. The preferred sites are the walls of the container (in the case of a liquid), inclusions, grain boundaries, stacking faults and dislocations (in the case of solids).

In Fig. 3.5, the nucleation of the new phase β on the surface of an inclusion δ embedded in the parent α phase is schematically shown. The β particle is in the form of a spherical cap. The contact angle θ is determined by equilibrium between surface tension forces (γ) acting at the periphery of the β particle. The energy of the inclusion surface is utilized during nucleation. The barrier for heterogeneous nucleation Δf^*_{het} turns out to be

$$\Delta f^*_{\text{het}} = \frac{1}{4} \Delta f^*_{\text{homo}} (2 - 3 \cos \theta + \cos^3 \theta) \quad (3.8)$$

The key to the reduction of the nucleation barrier is a small value of θ . When the crystal structures of δ and β are similar and the lattice parameters are nearly equal, a low energy interface forms between them. The contact angle θ is then small and the nucleation barrier is very effectively lowered.

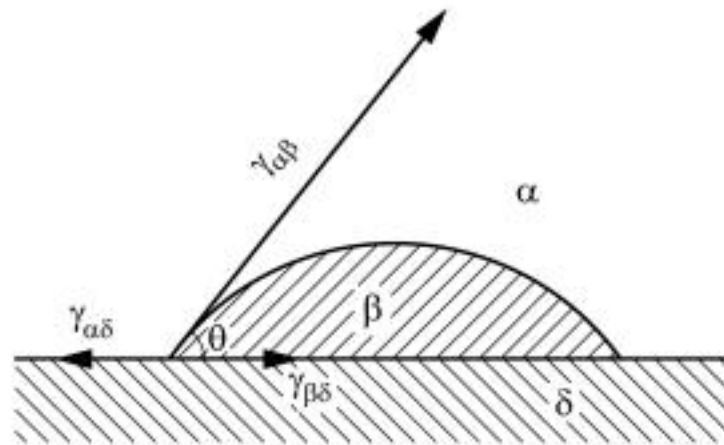


Figure 3.5 The nucleation of a β phase particle on the surface of a nucleating agent δ . The surface tension forces $\gamma_{\alpha\beta}$, etc. act along the indicated directions.

Overall transformation rate The overall transformation rate dX/dt , where X is the fraction of the product formed, is a function of both the nucleation rate and the growth rate. It has the same temperature dependence as nucleation and



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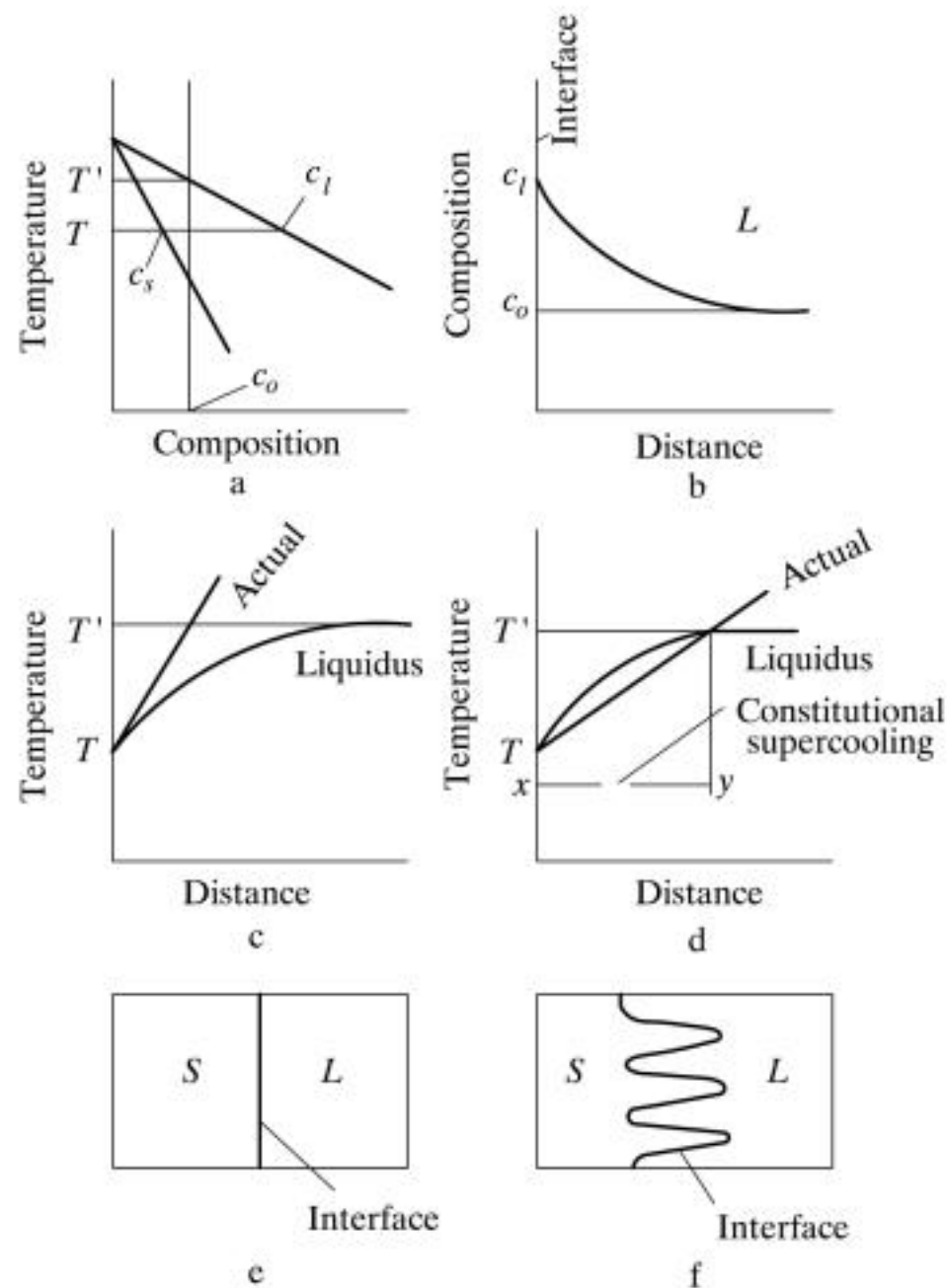


Figure 3.7 Constitutional supercooling: (a) the phase diagram, (b) the composition profile in the liquid ahead of the interface, (c) and (d) the temperature profiles ahead of the interface, and (e) and (f) the profiles of the interface.

Formation of dendrites Under conditions of constitutional supercooling, a cell structure with protrusions forms. If the velocity of the interface (the growth rate of the cells) increases further, the cells tend to grow along certain preferred directions such as $\langle 100 \rangle$ for cubic metals. Then a dendrite structure develops, where the growing crystal is in the form of a Christmas tree, Fig. 3.8. Recall that the $\langle 100 \rangle$ directions are three mutually perpendicular directions. One of them coincides with the growth direction of the central (primary) arm of the dendrite. The other two perpendicular $\langle 100 \rangle$ directions correspond to the secondary and tertiary arms of the dendrite. A number of primary dendrites might have branched off from the same nucleus and thus be part of the same grain, see Fig. 3.8. The spacing between dendrite arms decreases with increasing cooling rate.



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out of the ingot. The left-over CO is in the form of bubbles or wormholes between columnar grains. The turbulence during gas evolution in a rimmed ingot physically transports the metal to different parts, causing macrosegregation to a greater extent. *Capping* is a variation of rimming. After the ingot has rimmed for some time, a cast iron cap is placed on top of the mould to freeze the top and stop the rimming action.

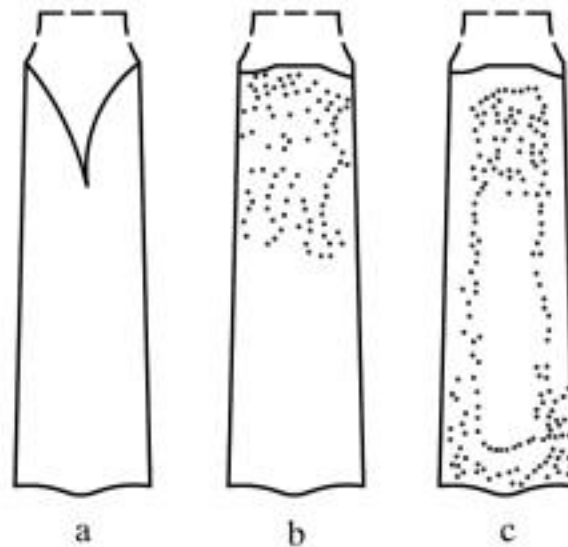


Figure 3.10 (a) Killed ingot, showing the shrinkage cavity at the top, (b) semikilled ingot, with CO bubbles near the top half, and (c) rimmed ingot with CO bubbles throughout the volume.

The upper part containing the exposed pipe in killed steels has to be rejected and this decreases the yield to about 80%. The yield from a rimmed ingot is higher.

Continuous casting Continuous casting is now an established method of producing semi-finished steel products like billets, blooms and slabs. In this method, the liquid steel is first poured from a ladle into a refractory-lined tundish, which directs the flow into a vertical water-cooled copper mould. The strand of steel which comes out of the mould is semi-solid, with a liquid core encased in a thick solidified shell. After solidification is complete, a welding torch cuts the strand into the desired lengths. The structure of the continuously-cast steel is similar to that of an ingot: a chill zone on the surface, followed by columnar grains and a central equiaxed zone.

Continuous casting offers several advantages. It yields a semi-finished shape (billet, bloom or slab), in contrast to the unfinished shape obtained in ingot casting. It provides better productivity and yield, reduced energy and manpower, improved product quality and consistency of quality, as compared to ingot casting. Only a killed steel can be continuously cast. The yield here is about 10% more, as there is no rejection in the form of the exposed pipe of an ingot. A rimmed steel cannot be continuously cast, as the rimming action can puncture holes in the solidified shell, with liquid pouring out.

Rapid solidification processing The cooling rate in rapid solidification processing (RSP) is in excess of 10^3 °C per second. In the *melt spinning* method,



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The interlamellar spacing If the steel is cooled slowly, the effective transformation temperature is just below the eutectoid temperature. This corresponds to a combination of a high growth rate and a relatively low nucleation rate, see Fig. 3.4. The crystals of ferrite and cementite are relatively coarse. If the steel is cooled more rapidly, the effective transformation temperature is lower and corresponds to a high nucleation rate and a relatively low growth rate. Note that carbon diffusion is required for growth and that the diffusion rate of carbon decreases with decreasing temperature. The crystals formed are, therefore, finer. The size of the crystals in pearlite is measured by a parameter called the interlamellar spacing.

The *interlamellar spacing* of a pearlite colony is defined as the distance from the centre of a ferrite (or cementite) plate to the centre of the next ferrite (or cementite) plate. The true spacing is seen under the microscope, when lamellae of a colony are perpendicular to the plane of observation. When they are inclined at an angle other than 90° , the measured spacing is larger than the true spacing. Experimentally, it is observed that the true interlamellar spacing is constant for a constant transformation temperature. It decreases with decreasing reaction temperature. In Fig. 3.15, the logarithm of interlamellar spacing in Å is plotted against the degree of supercooling ΔT below the eutectoid temperature. Transformation very near the eutectoid temperature yields coarse pearlite, with the interlamellar spacing of about $1\text{--}2\text{ }\mu\text{m}$. Coarse pearlite is soft, with a hardness of about $5\text{ }R_c$. The pearlite formed near the nose of the C-curve (550°C) is very fine, the spacing being less than $0.1\text{ }\mu\text{m}$. Hardness of very fine pearlite can be as high as $40\text{ }R_c$.

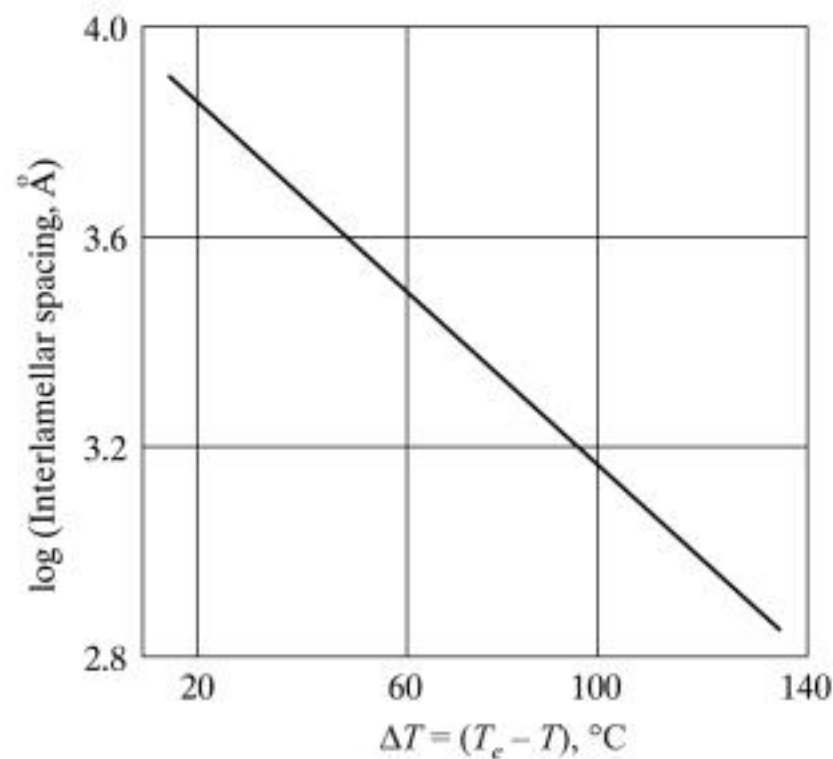


Figure 3.15 The variation of the interlamellar spacing with the degree of supercooling ΔT below the eutectoid temperature T_e .



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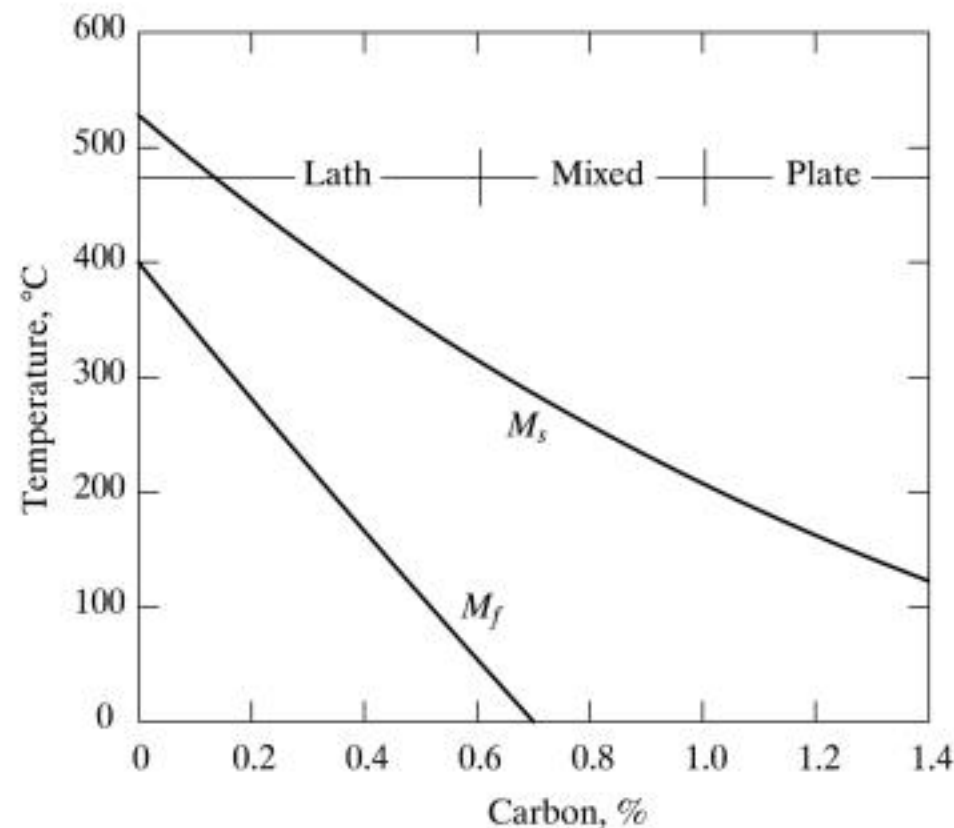


Figure 3.18 The effect of carbon on the M_s and M_f temperatures and on the morphology of martensite in plain carbon steels.

Other characteristics Two morphologies of martensite are commonly observed:

1. lath martensite, and
2. plate martensite.

At low concentrations of carbon and alloying elements and at high temperatures, lath martensite forms (see Fig. 3.19). The lath unit has the shape of a strip. A number of laths form in a parallel fashion, more or less completely filling austenite, Fig. 3.19a. At higher concentrations of carbon and alloying elements and at lower transformation temperatures, plate martensite forms. The plates are arranged in a non-parallel fashion, partitioning the austenite grains into several smaller pockets, Fig. 3.19b.

The hardness of martensite is a function of its carbon content, as shown in Fig. 3.20. The hardness values at 0.1 wt% C intervals are also listed below.

Carbon, wt.%	0.1	0.2	0.3	0.4	0.5	0.6	0.8
Hardness, R_c	38	44	50	57	60	63	65

To realize the full hardness, the steel should have at least 0.6% C. *Alloying elements have no significant effect on the hardness of martensite.* Martensite with hardness below 55 R_c has some amount of ductility. Those with hardness more than 60 R_c are generally brittle.



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externally applied stress above M_s (in place of a temperature change) can cause the same cyclic phenomenon, i.e. the plates appear and grow on increasing the applied stress; they shrink on decreasing the stress and disappear when the stress falls to zero. Some of the factors which favour the thermoelastic behaviour are a small hysteresis, a small shear strain associated with the martensitic transformation and accommodation of transformation strains *elastically*, without the occurrence of the irreversible plastic deformation.

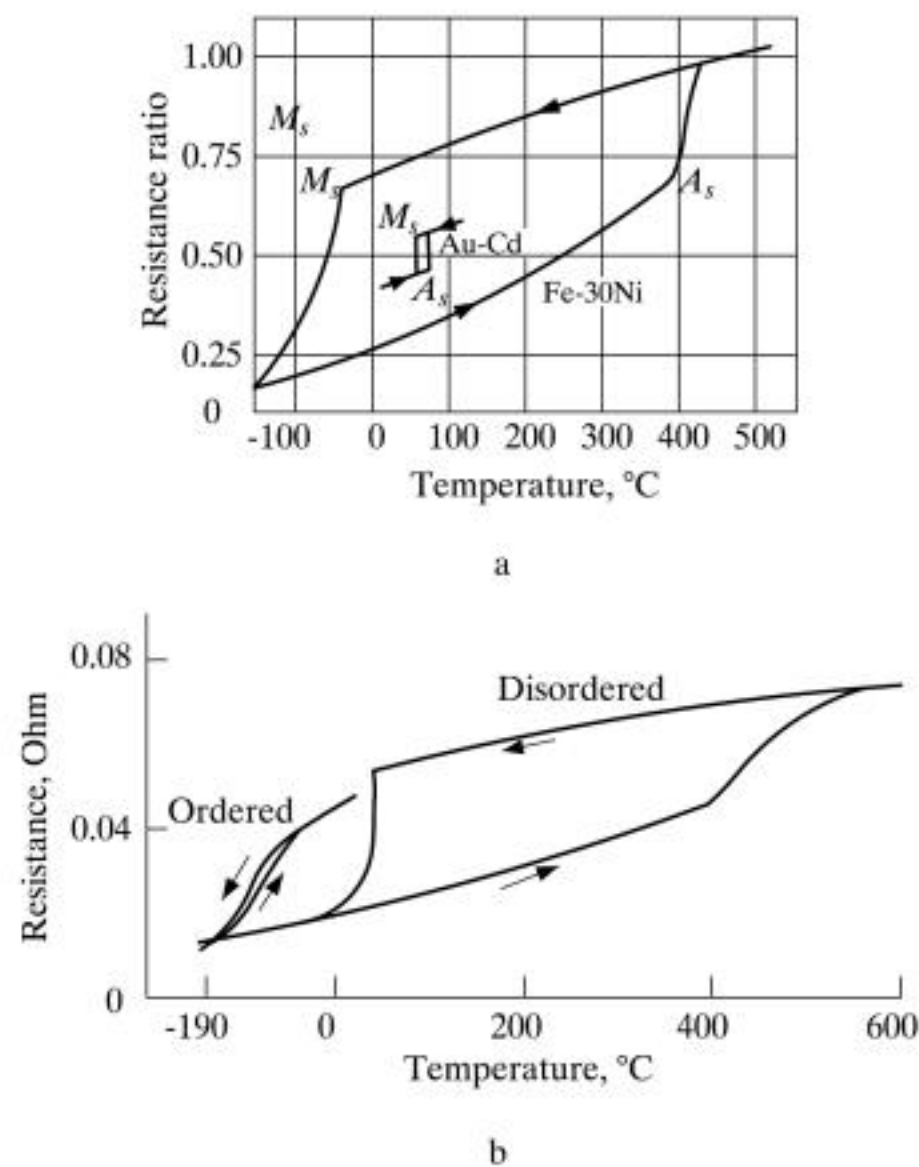


Figure 3.22 The reverse transformation of martensite to austenite starts at A_s .

Shape memory alloys The fact that the martensitic transformation in thermoelastic alloys can be brought about by thermal and/or mechanical means has led to the development of shape memory alloys. A well-known shape memory alloy NiTi (called *Nitinol*) has nearly equiatomic proportions of Ni and Ti. The stress-strain curve of nitinol at 10°C is compared with the normal elastic behaviour of stainless steel in Fig. 3.23. The stainless steel exhibits a usual linear elastic region as per Hooke's law. The elastic strain is less than half of a percent. The



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2. Sinha, A.K., *Physical Metallurgy Handbook*, McGraw-Hill, New York (2003).

QUESTIONS

1. In order to increase the diffusion penetration distance by a factor of two, by how much should the diffusion time be changed and why?
2. Why is a nickel steel heated in air prone to decarburization and not to denickelization?
3. Derive the results given in Eqs. (3.6) and (3.7).
4. Explain how the grain size is determined by the nucleation and growth rates.
5. List the methods available to you for refining the grain size of a casting.
6. Distinguish between:
 - (i) microsegregation and macrosegregation, and
 - (ii) normal segregation and inverse segregation.
7. Compare the characteristics of pearlite, bainite and martensite transformations in a tabular form.
8. Arrange the following microconstituents in order of decreasing free energy: pearlite, martensite, tempered martensite, ferrite and graphite, spheroidized cementite and ferrite.
9. What is the purpose of tempering a quenched steel? How can the tempered properties be optimized? Discuss with examples.
10. Duralumin alloy rivets used in aircraft construction are kept at -40°C after the solution treatment. Whenever required, they are taken out and riveted on the spot. Explain the reason for keeping them at -40°C and how the riveting operation at room temperature strengthens the rivet. (*Hint*: Strain-induced precipitation is faster than normal precipitation).
11. Very fine alloy carbide particles effectively pin down the austenite grain boundaries during recrystallization. Explain how this results in grain refinement.



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martensite. The reason for the shift of the C-curve to the left is that the proeutectoid ferrite provides sites for pearlite nucleation and enhances its rate of formation.

A hypereutectoid steel has a similar T-T-T diagram, Fig. 4.2b. The upper limiting temperature here is the A_{cm} temperature and the proeutectoid phase is cementite. The shift of the nose to the left occurs here as well.

Effect of alloying elements *All common alloying elements (except cobalt) shift the nose of the C-curve to the right, thus making it easier to quench the steel past the nose without any transformation on the way. They are said to increase the hardenability of the steel. Also, in a number of cases, in the presence of an alloying element, separation of the pearlitic and the bainitic noses occurs. Figure 4.3 is the T-T-T diagram for a Ni-Cr-Mo low alloy steel with 0.4%C, 1.8%Ni, 0.8%Cr and 0.3%Mo. The pearlite nose is at 650°C and the nose time is ~100 s. The bainite nose is at 450°C and at ~10 s. The shorter time at the bainitic nose determines the hardenability of the steel. This time is still an order of magnitude larger than the nose time for a eutectoid steel.*

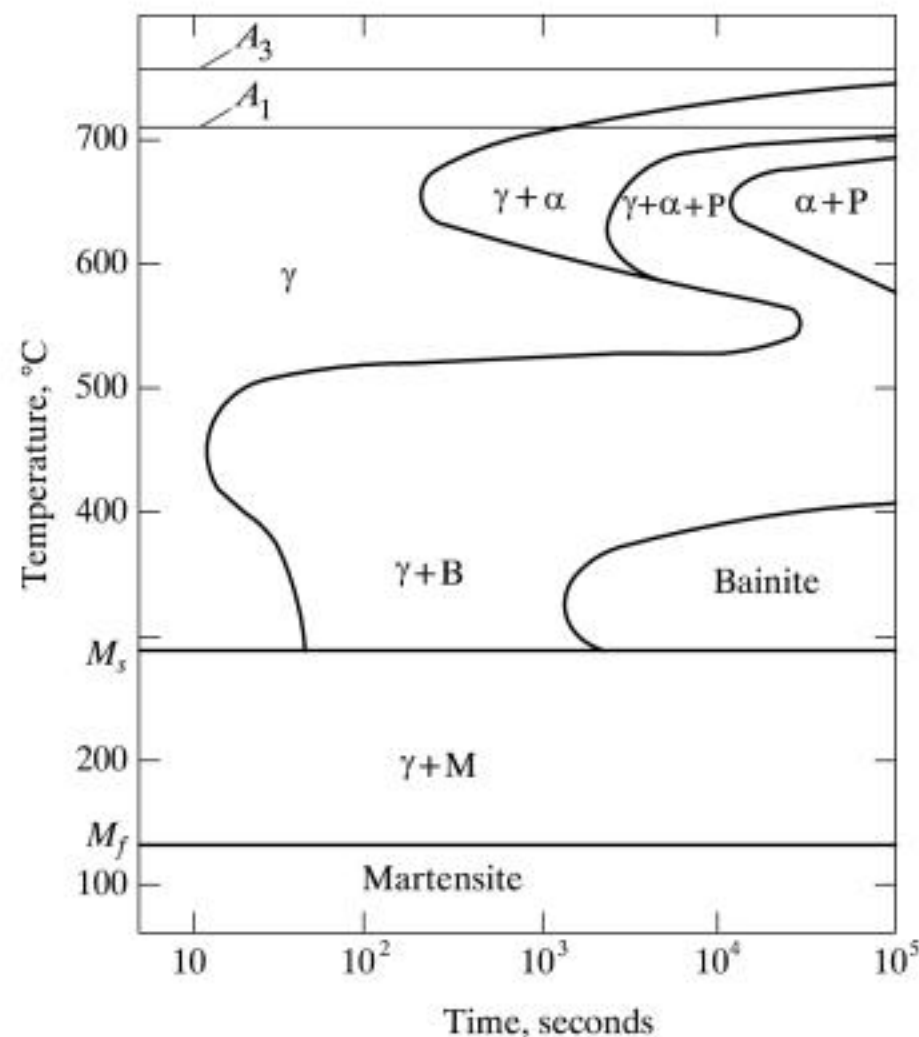


Figure 4.3 The T-T-T diagram for a low-alloy Ni-Cr-Mo steel, depicting two separate C-curves for the pearlitic and the bainitic transformations.



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Chromium and nickel in solution in ferrite or austenite improve the corrosion resistance of iron remarkably. Stainless steels, which contain chromium with or without nickel, are discussed in Chaps. 6 and 7.

An important function of alloying elements in steels is to *increase the hardenability*, by shifting the nose of the C-curve to the right on the T-T-T diagram. *To be effective, the alloying element must be in solution in austenite.*

Second phase particles have both desirable and undesirable effects in steels. They are the initiators of voids during deformation, which coalesce ultimately to cause fracture. Silicates and sulphides get elongated during hot rolling, which result in poor toughness, when tested transverse to the direction of rolling. MnS removes *hot shortness* (brittleness at elevated temperatures). Sulphides, phosphides and insoluble metals improve the *machinability* of steels.

Fine spherical particles of alloy carbides pin down the migrating boundaries during recrystallization and grain growth. Thus they help in maintaining a fine grain size. In the presence of alloy carbides, the abrasion and wear resistance of steels is improved. Some alloy carbides such as MoC and VC enhance the high temperature (creep) strength, by providing a fine stable dispersion of particles in the matrix.

The order of increasing stability of carbides in a steel is as follows:

Ni	Si	Al	Fe	Mn	Cr	W	Mo	V	Ti	Nb
⏟										
noncarbide formers			increasing carbide stability —————→							

When nitrogen is present, many of the carbide formers listed above form carbonitrides or nitrides, which are also very stable and abrasion resistant.

4.4 FORMATION OF AUSTENITE

In many heat treatment processes, the first step consists of heating the steel into the austenitic range, the process known as *austenitization*. Austenite starts to form when the steel is heated above A_{c1} . The kinetics of its formation depends to some extent on the initial microstructure. With tempered martensite as the initial structure, austenitization occurs more rapidly than with coarse pearlite.

The rate of austenitization can be shown on a T-T-T diagram, Fig. 4.6. The carbon content is 0.8% and the initial microstructure is fine pearlite. The rate of austenite formation increases rapidly with temperature. For example, full austenitization occurs in less than 5 s at 800°C. However, some residual carbides are left over, as cementite dissolves more slowly than ferrite. The third curve on the right in Fig. 4.6 indicates the completion of *cementite dissolution*. Even after this, the austenite is not homogeneous, as it has less carbon in regions which were initially ferrite and more carbon in regions of initial cementite. The fourth and the last curve on the right shows the *homogenization* of austenite. It is seen that the time for complete homogenization is much longer than for the formation of 99% of austenite (second curve in Fig. 4.6)



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at the centre will also exceed the critical cooling rate and the steel will have hardened throughout the cross-section. However, drastic quenching causes other undesirable effects such as *warping and cracking* of the steel. *The hardenability of a steel is defined as its ability to harden (by forming martensite) throughout its cross-section, without having to resort to drastic quenching.* Alloying elements play a crucial role in increasing the hardenability. If this were not so, we will be building machine components as layer structures, where each layer is thin enough to harden fully!

Grossman's critical diameter In Grossman's experiments, a number of steel rods of different diameters, Fig. 4.9, are austenitized and quenched in the same bath. To avoid end effects, the length of the rod must be at least five times the diameter. The smallest diameter rod hardens right through. If $C \geq 0.6\%$, a hardness of $\sim 65 R_c$ is obtained across the entire cross-section. As the diameter of the rod increases, the cooling rate at the centre decreases and becomes less than the critical cooling rate. Then, the hardness falls short of the full value of $65 R_c$.

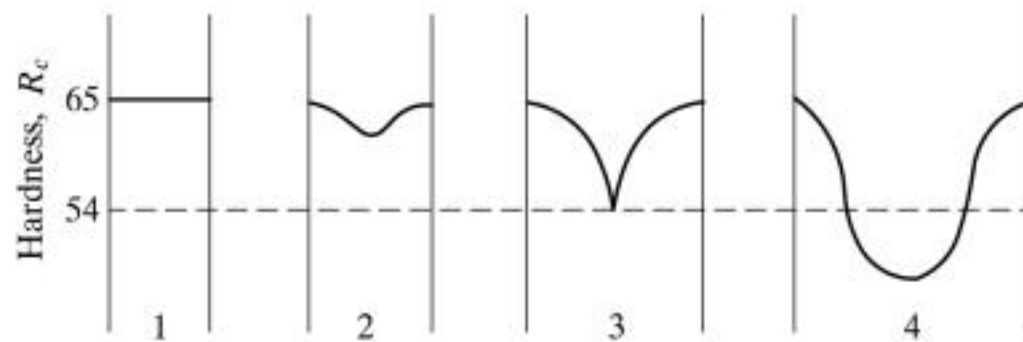


Figure 4.9 Rods of different diameters 1, 2, 3, 4 harden to different depths on quenching in the same medium. The critical diameter is that of rod 3.

The largest diameter rod 4 in Fig. 4.9 has only fine pearlite at the centre. The hardness changes most rapidly at a value of $54 R_c$, corresponding to a microstructure of 50% martensite–50% pearlite. *This inflection point is taken as the transition zone between the hardened and not-hardened parts of the cross-section.* For a certain diameter (3) less than the largest, the microstructure at the centre is 50% martensite–50% pearlite with a hardness of $54 R_c$. This diameter is called the *critical diameter*. Diameters more than the critical will not harden throughout, whereas smaller diameters will. The critical diameter is thus a measure of the hardenability of a steel.

Severity of quenching media If the critical diameter is to be defined as a material property that depends only on its composition and the austenitizing treatment, the effect of the quenching medium used should be eliminated. For this purpose, Grossman defined an ideal quenching medium and an ideal critical diameter that corresponds to quenching in that medium. The severity of quenching of a medium is defined by the parameter H .

$$H = \frac{\text{heat transfer coefficient between steel and medium}}{\text{thermal conductivity of the steel}}$$



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simultaneously across the section. Residual stresses induced are a minimum. The steel is tempered in the usual way to the desired hardness. As the cooling rate through the martensitic range is less here, more austenite tends to be retained due to thermal stabilization.

Figure 4.17c illustrates the *austempering* process. Here, the steel is quenched into the bainitic bay above M_s and kept isothermally till all the austenite is converted to lower bainite.

The pearlitic nose must be further to the right for austempering and martempering than for conventional quenching, to prevent any transformation near the nose. Also, for martempering, the time at the bainitic bay should be long enough for temperature equalization before any transformation occurs.

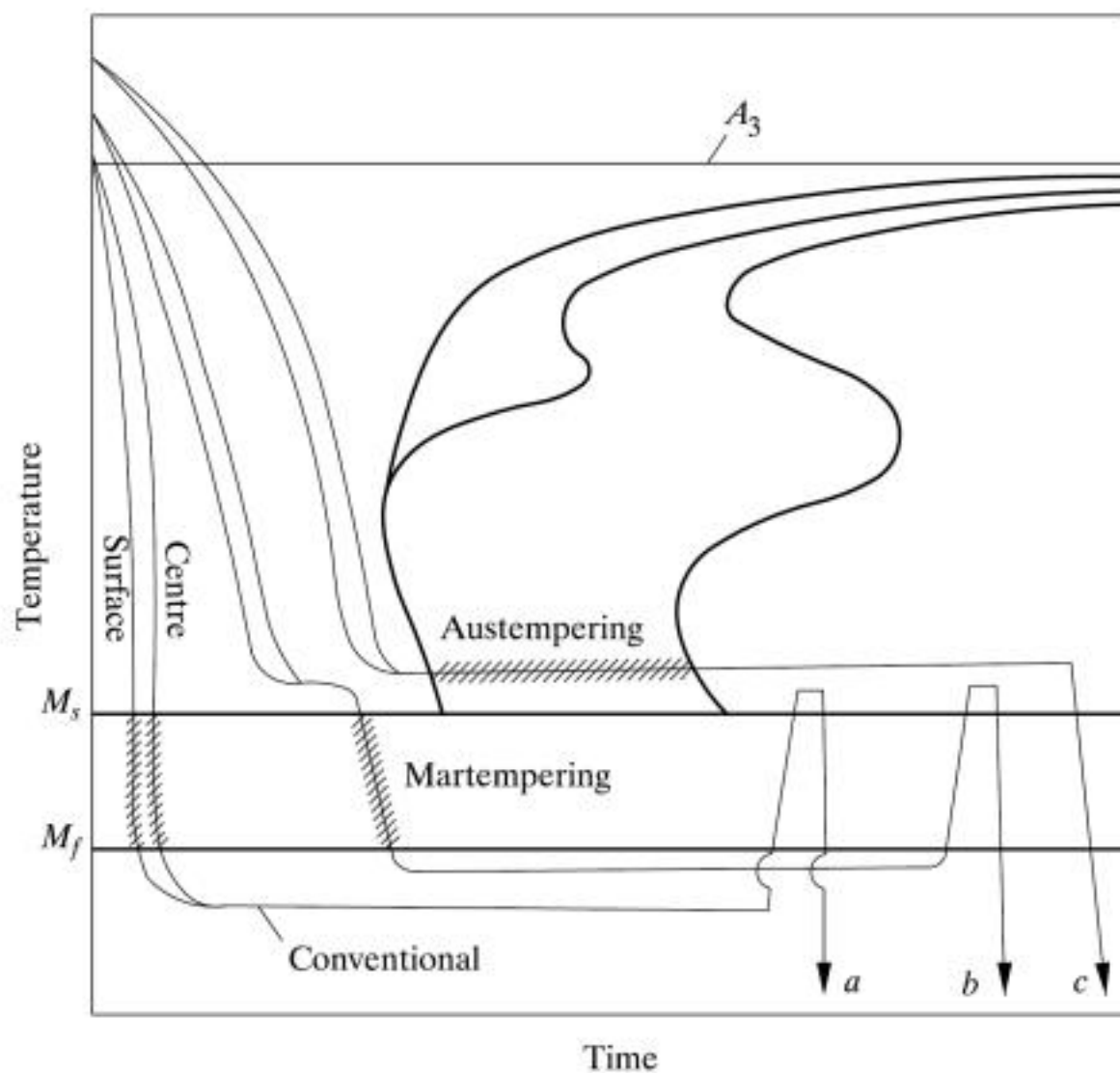


Figure 4.17 Heat treatment processes to reduce quenching stresses: (a) conventional quenching and tempering, (b) martempering, and (c) austempering.

The impact energy and elongation of a 0.95%C steel heat treated to the same final hardness by these three methods are compared as follows:



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